MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGISTER, DIGITAL, KEY CODE PJB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic register module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665		Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.803 failures per million operating hours for class I or 6.062 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PJB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/317-1 for class I and M28787/317-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 V dc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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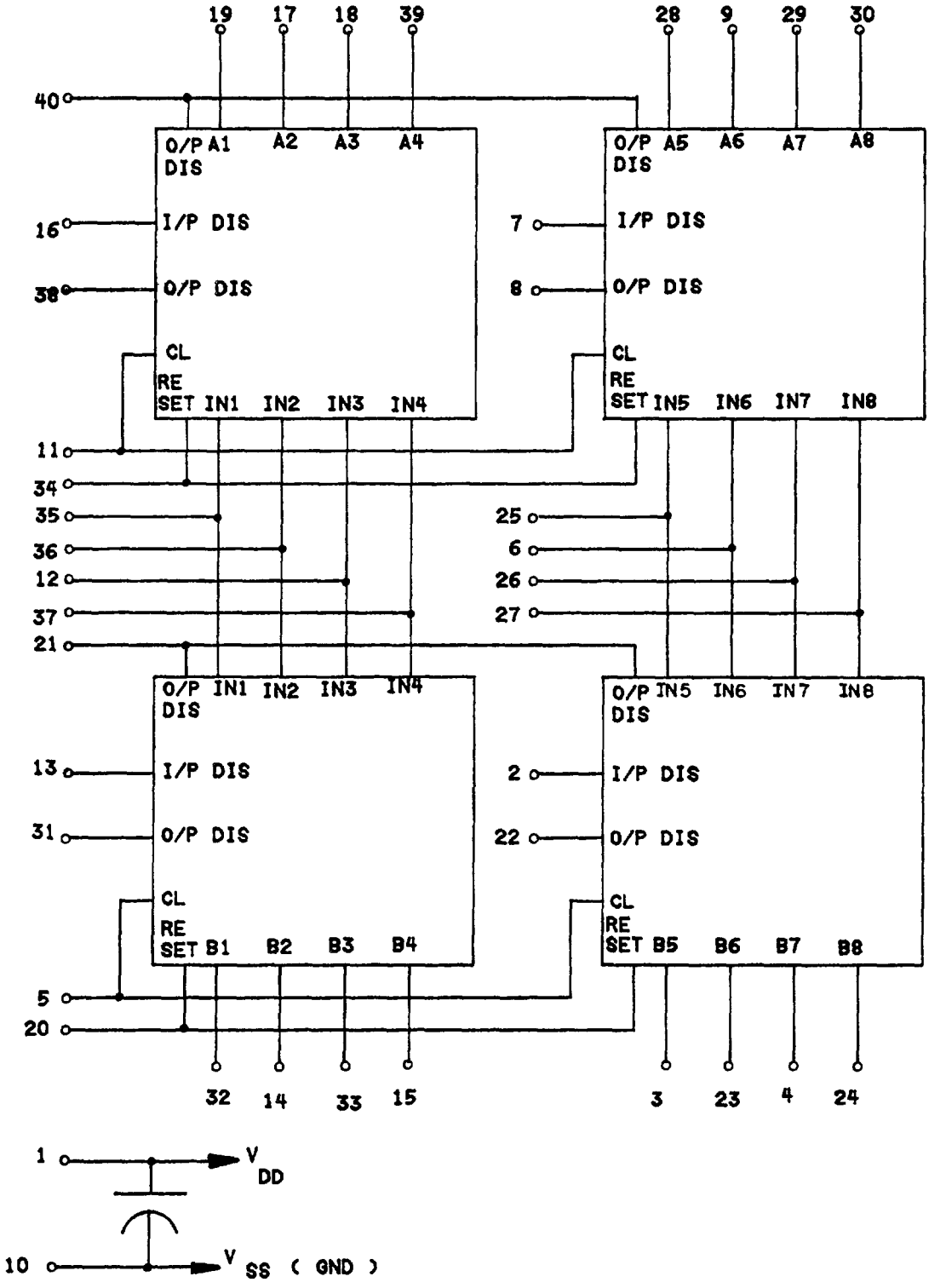


FIGURE 1. PJB module function diagram, register.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{DD} stress	IV	Q	← Module must pass all remaining tests →										
C1-C22	Input clamp diode voltage, negative	V	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
D1-D22	Input clamp diode voltage, positive	VI	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
E1-E7	Functional	VII	X	← See table VII →										
F1-F16	Output voltage, low level (I_{OL} = 0)	VIII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
G1-G16	Output voltage, high level (I_{OH} = 0)	IX	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
H1-H16	Output voltage, low level (V_{DD} = 5 V)	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J16	Output voltage, high level (V_{DD} = 5 V)	XI	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
K1-K16	Output voltage, low level (V_{DD} = 15 V)	XII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L16	Output voltage, high level (V_{DD} = 15 V)	XIII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
M1-M22	Input current, low level	XIV	X	0	-200	---	---	0	-1000	---	---	0	-1000	nA
N1-N22	Input current, high level	XV	X	0	200	---	---	0	1000	---	---	0	1000	nA
P1-P16	Output leakage current, low level (I_{OZL})	XVI	X	0.0	-0.4	---	---	0.0	-12.0	---	---	0.0	-12.0	µA
Q1-Q16	Output leakage current, high level (I_{OZH})	XVII	X	0.0	0.4	---	---	0.0	12.0	---	---	0.0	12.0	µA

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1, R2	Power supply current	XVIII	X	0	80	0	80	0	1400	0	80	0	2400	µA
S1-S16	Transition time (t_{TLH}), V_{DD} = 5 V, C_L = 50 pF	XIX	S	36	360	36	360	38	385	30	302	41	400	ns
S17-S32	Transition time (t_{TLH}), V_{DD} = 5 V, C_L = 150 pF	XIX	Q	84	840	84	840	91	916	74	743	100	960	ns
T1-T16	Transition time (t_{THL}), V_{DD} = 5 V, C_L = 50 pF	XX	S	20	200	20	200	21	213	17	171	22	218	ns
T17-T32	Transition time (t_{THL}), V_{DD} = 5 V, C_L = 150 pF	XX	Q	44	440	44	440	47	478	35	354	52	500	ns
U1-U16	Transition time (t_{TLH}), V_{DD} = 10 V, C_L = 50 pF	XXI	Q	18	180	18	180	19	191	15	156	20	198	ns
V1-V16	Transition time (t_{THL}), V_{DD} = 10 V, C_L = 50 pF	XXII	Q	10	100	10	100	10	105	8	88	11	108	ns
W1-W16	Transition time (t_{TLH}), V_{DD} = 15 V, C_L = 50 pF	XXIII	Q	13	130	13	130	13	138	11	112	14	143	ns
X1-X16	Transition time (t_{THL}), V_{DD} = 15 V, C_L = 50 pF	XXIV	Q	8	80	8	80	8	84	7	70	8	87	ns
Y1-Y16	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 50 pF, t_{PLH}	XXV	S	60	600	60	600	61	611	57	576	62	618	ns
Y17-Y32	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 50 pF, t_{PHL}	XXV	S	60	600	60	600	61	611	57	576	62	618	ns
Z1-Z16	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 150 pF, t_{PLH}	XXVI	Q	80	800	80	800	83	832	72	728	86	850	ns
Z17-Z32	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 150 pF, t_{PHL}	XXVI	Q	80	800	80	800	83	832	72	728	86	850	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1-AA16	Propagation delay time (CL to output), V_{DD} = 10 V, C_L = 50 pF, t_{PLH}	XXVII	Q	25	250	25	250	25	255	23	239	26	258	ns
AA17-AA32	Propagation delay time (CL to output), V_{DD} = 10 V, C_L = 50 pF, t_{PHL}	XXVII	Q	25	250	25	250	25	255	23	239	26	258	ns
BB1-BB16	Propagation delay time (CL to output), V_{DD} = 15 V, C_L = 50 pF, t_{PLH}	XXVIII	Q	18	180	18	180	18	184	17	170	18	187	ns
BB17-BB32	Propagation delay time (CL to output), V_{DD} = 15 V, C_L = 50 pF, t_{PHL}	XXVIII	Q	18	180	18	180	18	184	17	170	18	187	ns
CC1-CC16	Propagation delay time (Reset to output), V_{DD} = 5 V, C_L = 50 pF, t_{PHL}	XXIX	S	60	600	60	600	61	611	57	576	62	618	ns
CC17-CC32	Propagation delay time (Reset to output), V_{DD} = 5 V, C_L = 150 pF, t_{PHL}	XXIX	Q	80	800	80	800	83	832	72	728	86	850	ns
DD1-DD16	Propagation delay time (Reset to output), V_{DD} = 10 V, C_L = 50 pF, t_{PHL}	XXX	Q	25	250	25	250	25	255	23	239	26	258	ns
EE1-EE16	Propagation delay time (Reset to output), V_{DD} = 15 V, C_L = 50 pF, t_{PHL}	XXXI	Q	18	180	18	180	18	184	17	170	18	187	ns
FF1-FF16	Disable delay time (t_{PLZ}), V_{DD} = 5 V, C_L = 50 pF,	XXXII	S	70	700	70	700	74	740	70	700	80	770	ns
FF17-FF32	Enable delay time (t_{PZL}), V_{DD} = 5 V, C_L = 50 pF,	XXXII	S	17	170	17	170	19	190	17	170	22	205	ns
FF33-FF48	Disable delay time (t_{PHZ}), V_{DD} = 5 V, C_L = 50 pF,	XXXII	S	0.15	1.50	0.15	1.50	0.16	1.60	0.15	1.50	0.17	1.65	µs

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See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF49-FF64	Enable delay time (t_{PZH}), V_{DD} = 5 V, C_L = 50 pF	XXXII	S	14	140	14	140	16	160	14	140	19	175	ns
GG1-GG16	Disable delay time (t_{PLZ}), V_{DD} = 10 V, C_L = 50 pF	XXXIII	Q	66	660	66	660	70	700	66	660	76	730	ns
GG17-GG32	Enable delay time (t_{PZL}), V_{DD} = 10 V, C_L = 50 pF	XXXIII	Q	6	60	6	60	8	75	6	60	10	85	ns
GG33-GG48	Disable delay time (t_{PHZ}), V_{DD} = 10 V, C_L = 50 pF	XXXIII	Q	0.15	1.50	0.15	1.50	0.16	1.60	0.15	1.50	0.17	1.65	µs
GG49-GG64	Enable delay time (t_{PZH}), V_{DD} = 10 V, C_L = 50 pF	XXXIII	Q	6	60	6	60	8	75	6	60	10	85	ns
HH1-HH16	Disable delay time (t_{PLZ}), V_{DD} = 15 V, C_L = 50 pF	XXXIV	Q	62	620	62	620	66	660	62	620	72	690	ns
HH17-HH32	Enable delay time (t_{PZL}), V_{DD} = 15 V, C_L = 50 pF	XXXIV	Q	5	50	5	50	7	65	5	50	9	75	ns
HH33-HH48	Disable delay time (t_{PHZ}), V_{DD} = 15 V, C_L = 50 pF	XXXIV	Q	0.14	1.40	0.14	1.40	0.15	1.50	0.14	1.40	0.16	1.55	µs
HH49-HH64	Enable delay time (t_{PZH}), V_{DD} = 15 V, C_L = 50 pF	XXXIV	Q	5	45	5	45	6	60	5	45	8	70	ns
JJ1-JJ16	Hold time $t_{H(H)}$, V_{DD} = 5 V, C_L = 50 pF	XXXV	S	3	5	3	5	3	5	3	5	3	5	µs
JJ17-JJ32	Hold time $t_{H(L)}$, V_{DD} = 5 V, C_L = 50 pF	XXXV	S	3	5	3	5	3	5	3	5	3	5	µs
KK1-KK16	Setup time $t_{S(H)}$, V_{DD} = 5 V, C_L = 50 pF	XXXVI	S	3	5	3	5	3	5	3	5	3	5	µs
KK17-KK32	Setup time $t_{S(L)}$, V_{DD} = 5 V, C_L = 50 pF	XXXVI	S	3	5	3	5	3	5	3	5	3	5	µs
LL1-LL16	Hold time $t_{H(H)}$, V_{DD} = 10 V, C_L = 50 pF	XXXVII	Q	3	5	3	5	3	5	3	5	3	5	µs

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
LL17-LL32	Hold time $t_{H(L)}$, V_{DD} = 10 V, C_L = 50 pF	XXXVII	Q	3	5	3	5	3	5	3	5	3	5	µs
MM1-MM16	Setup time $t_{S(H)}$, V_{DD} = 10 V, C_L = 50 pF	XXXVIII	Q	3	5	3	5	3	5	3	5	3	5	µs
MM17-MM32	Setup time $t_{S(L)}$, V_{DD} = 10 V, C_L = 50 pF	XXXVIII	Q	3	5	3	5	3	5	3	5	3	5	µs
NN1-NN16	Hold time $t_{H(H)}$, V_{DD} = 15 V, C_L = 50 pF	XXXIX	Q	3	5	3	5	3	5	3	5	3	5	µs
NN17-NN32	Hold time $t_{H(L)}$, V_{DD} = 15 V, C_L = 50 pF	XXXIX	Q	3	5	3	5	3	5	3	5	3	5	µs
PP1-PP16	Setup time $t_{S(H)}$, V_{DD} = 15 V, C_L = 50 pF	XL	Q	3	5	3	5	3	5	3	5	3	5	µs
PP17-PP32	Setup time $t_{S(L)}$, V_{DD} = 15 V, C_L = 50 pF	XL	Q	3	5	3	5	3	5	3	5	3	5	µs
QQ1-QQ16	Maximum shift frequency check	XLI	S	950	1150	950	1150	950	1150	950	1150	950	1150	ns
RR	Power supply filter capacitance	XLII	S	70	130	70	130	70	130	70	130	70	130	µF
SS1-SS8	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF
SS9-SS16	Input capacitance	XLIII	Q	2	25	2	25	2	25	2	25	2	25	pF
SS17-SS22	Input capacitance	XLIII	Q	2	50	2	50	2	50	2	50	2	50	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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* 4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through SS in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through RR shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition RR, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following life test and while still at the maximum class temperature, electrical tests E through RR shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through SS shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

		Class I	Class II	Unit
Maximum power dissipation		2105	2105	mW
Input voltage				
Low level:	V_{DD} = 5V	1.5 max	1.5 max	V
	V_{DD} = 10V	3.0 max	3.0 max	V
	V_{DD} = 15V	4.0 max	4.0 max	V
High level	V_{DD} = 5V	3.5 min	3.5 min	V
	V_{DD} = 10V	7.0 min	7.0 min	V
	V_{DD} = 15V	11.0 min	11.0 min	V
Output drive current				
Low level (V_{OL} = 0.4V)				
At -55°C		0.64	0.64	mA
At 0° and 25°C		0.51	0.51	mA
At 60°C		0.44	0.44	mA
At 85°C		0.36	0.36	mA
High level (V_{OH} = 4.6V)				
At -55°C		-0.25	-0.25	mA
At 0° and 25°C		-0.20	-0.20	mA
At 60°C		-0.17	-0.17	mA
At 85°C		-0.14	-0.14	mA
Setup and hold time				
T_{SET}				
V_{DD} = 5V		200	200	ns
V_{DD} = 10V		80	80	ns
V_{DD} = 15V		60	60	ns
T_{HOLD}				
V_{DD} = 5V		130	130	ns
V_{DD} = 10V		60	60	ns
V_{DD} = 15V		50	50	ns

6.5 Module function. The digital register module contains four 4-bit three state latches. Each latch contains independent input disable lines and output disable with provisions made for common output disable lines between two latches. Clock inputs, reset inputs, and data inputs are commoned between two latches.

6.5.1 Input to output relationship. Each latch shall have the input to output relationship of table II.

TABLE II. Input to output relationship.

Input					Output
Clock	I/P DIS	O/P DIS	Reset	Data In	Q
X	X	L	H	X	L
L	X	L	L	X	Qn
↑	H	L	L	X	Qn
↑	L	L	L	H	H
↑	L	L	L	L	L
H	L	L	L	X	Qn
X	X	H	X	X	H_{IZ}

L = Low level.
H = High level.
↑ = Positive transition of clock.
X = Don't care.
Qn = No change.
H_{IZ} = High impedance state.

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6.5.2 Paralleling of outputs. Paralleling of outputs to minimize bus lines is permitted only for the case where no two outputs connected to the same bus line are enabled simultaneously. Paralleling of outputs shall not be attempted for any other situation.

6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}2f)\ N + \text{Quiescent power dissipation.}$$

Where: C = load capacitance.
V_{DD} = Supply voltage.
f = operating frequency.
N = number of outputs.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

e. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

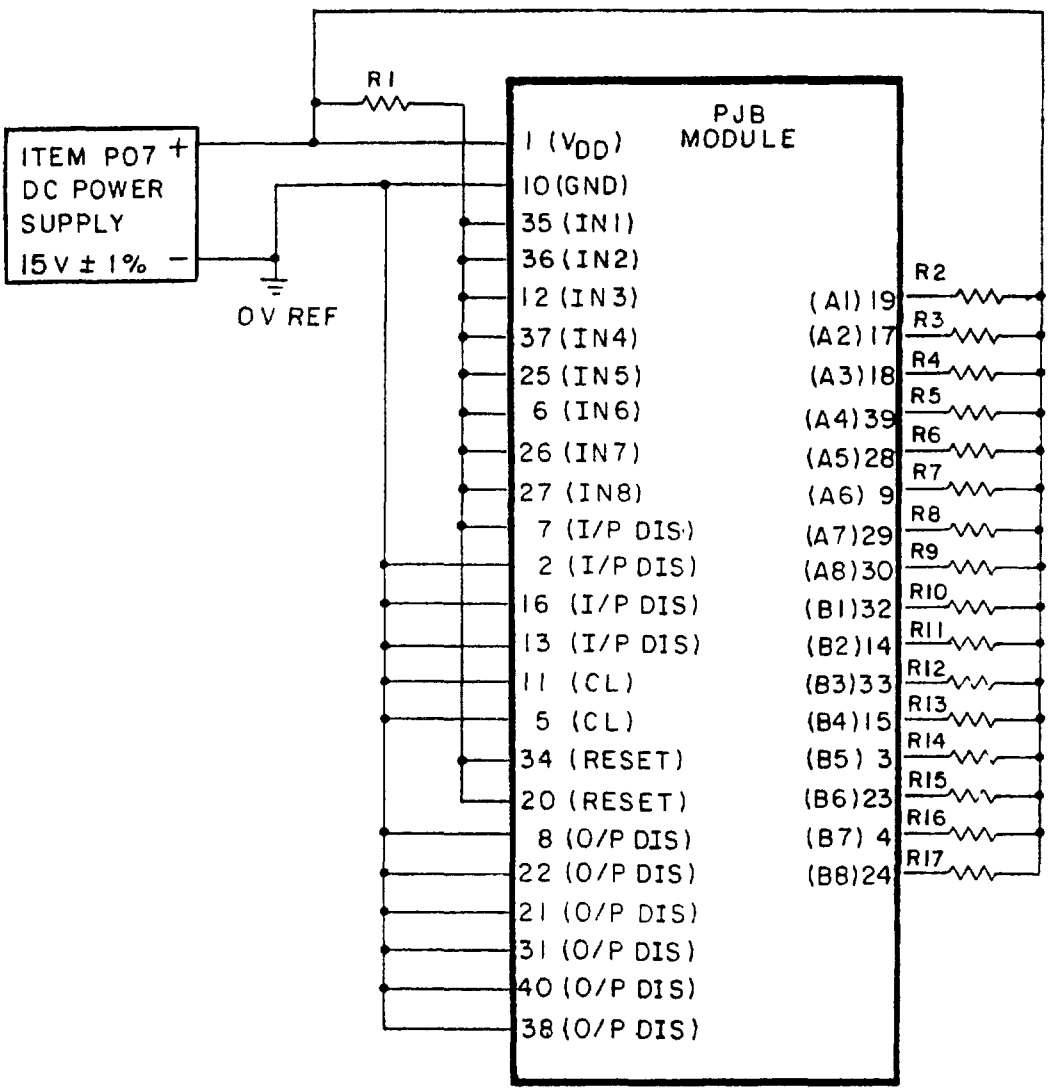
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
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Preparing activity:
Navy - EC

(Project 5963-N459)

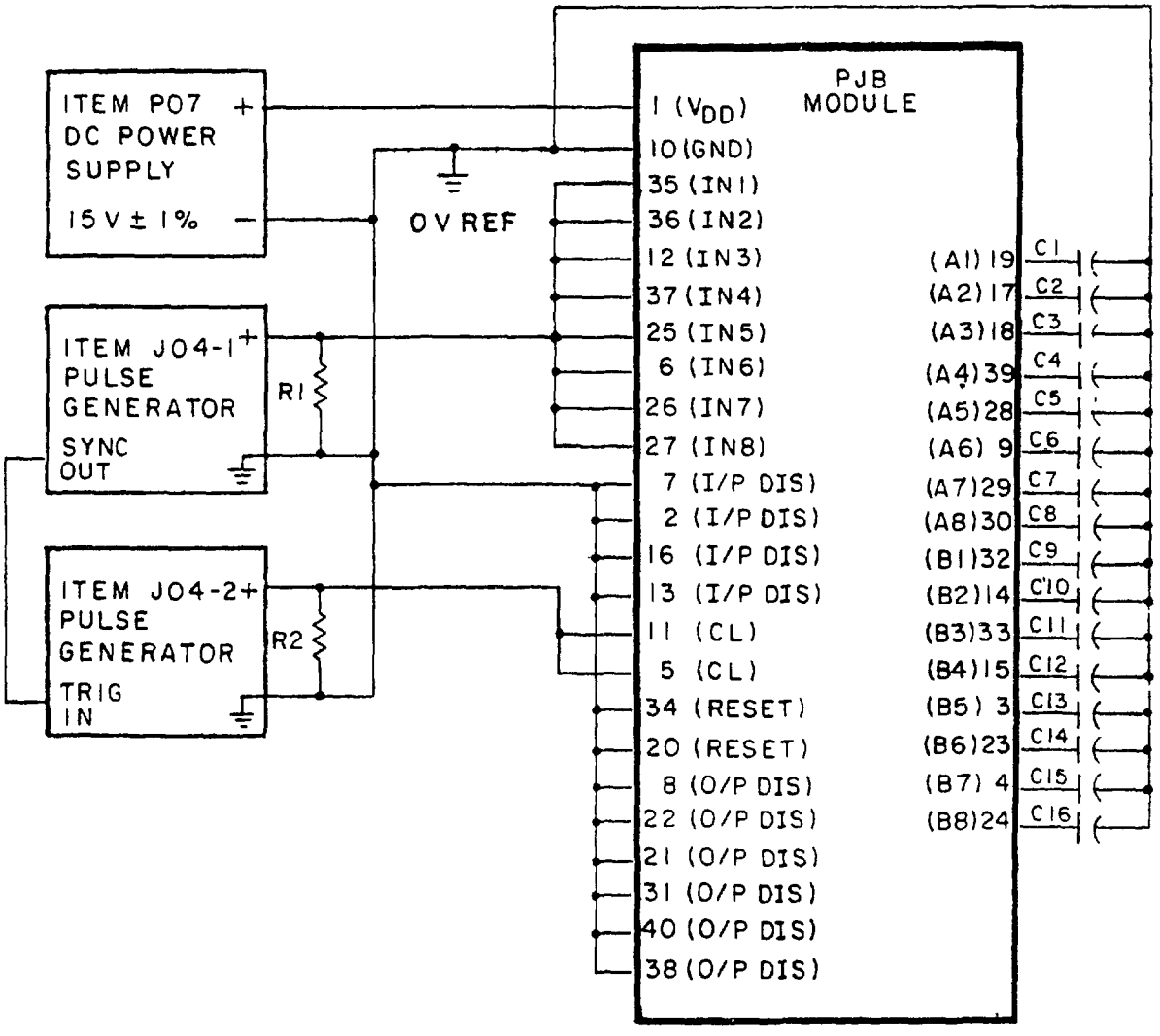
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NOTES.
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2-R17 = 11.5 kΩ ±1%, 1/8 W.

* FIGURE 2. Operating temperature and life test circuit.

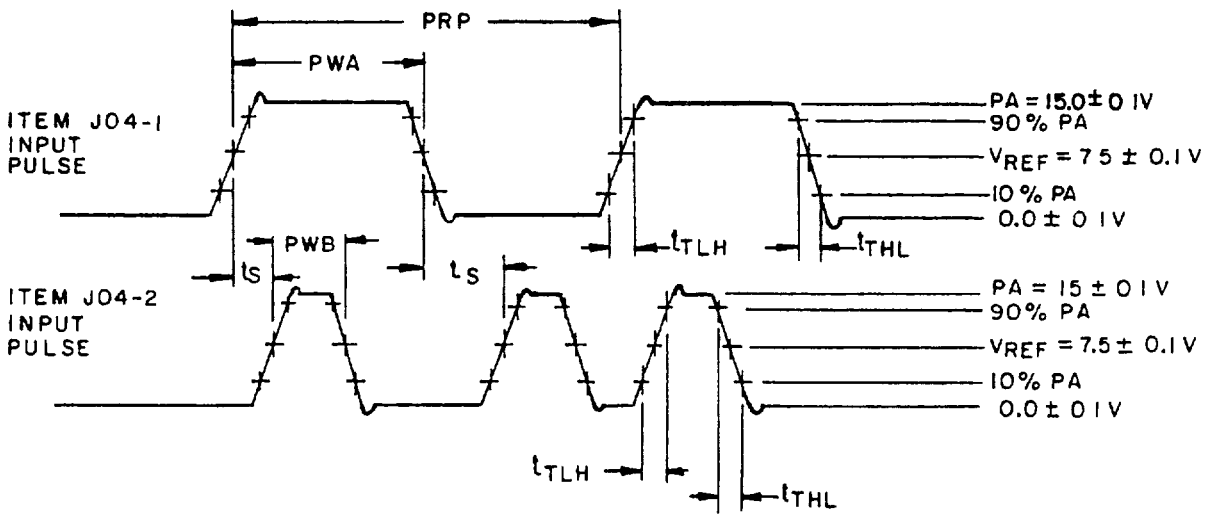
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NOTES:
1. R1 = R2 = 600Ω ±1%, 1/2 W.
2. C1-C16 = 50 pF ±10%, 20 V dc.

FIGURE 3. Component temperature test circuit.

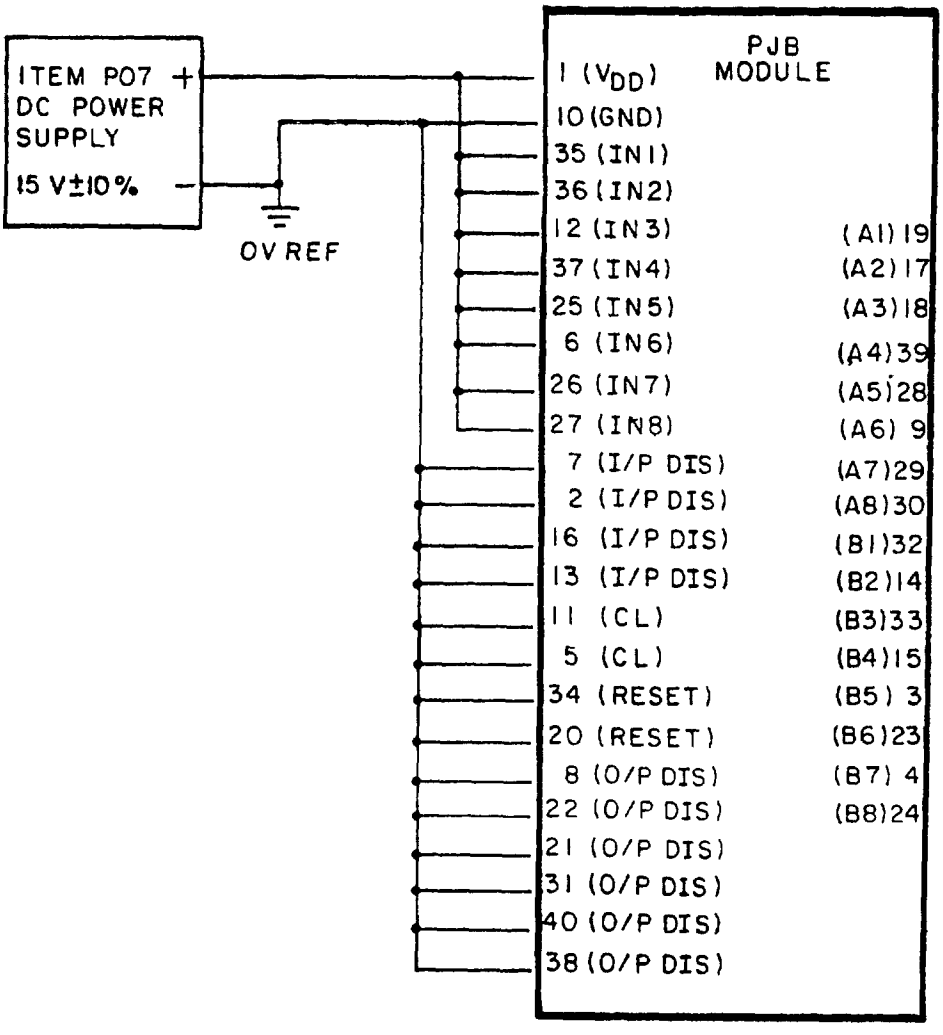
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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.
PWB = 250 ±25 ns.
t_S = 100 ±10 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 4. Component temperature test circuit waveforms.

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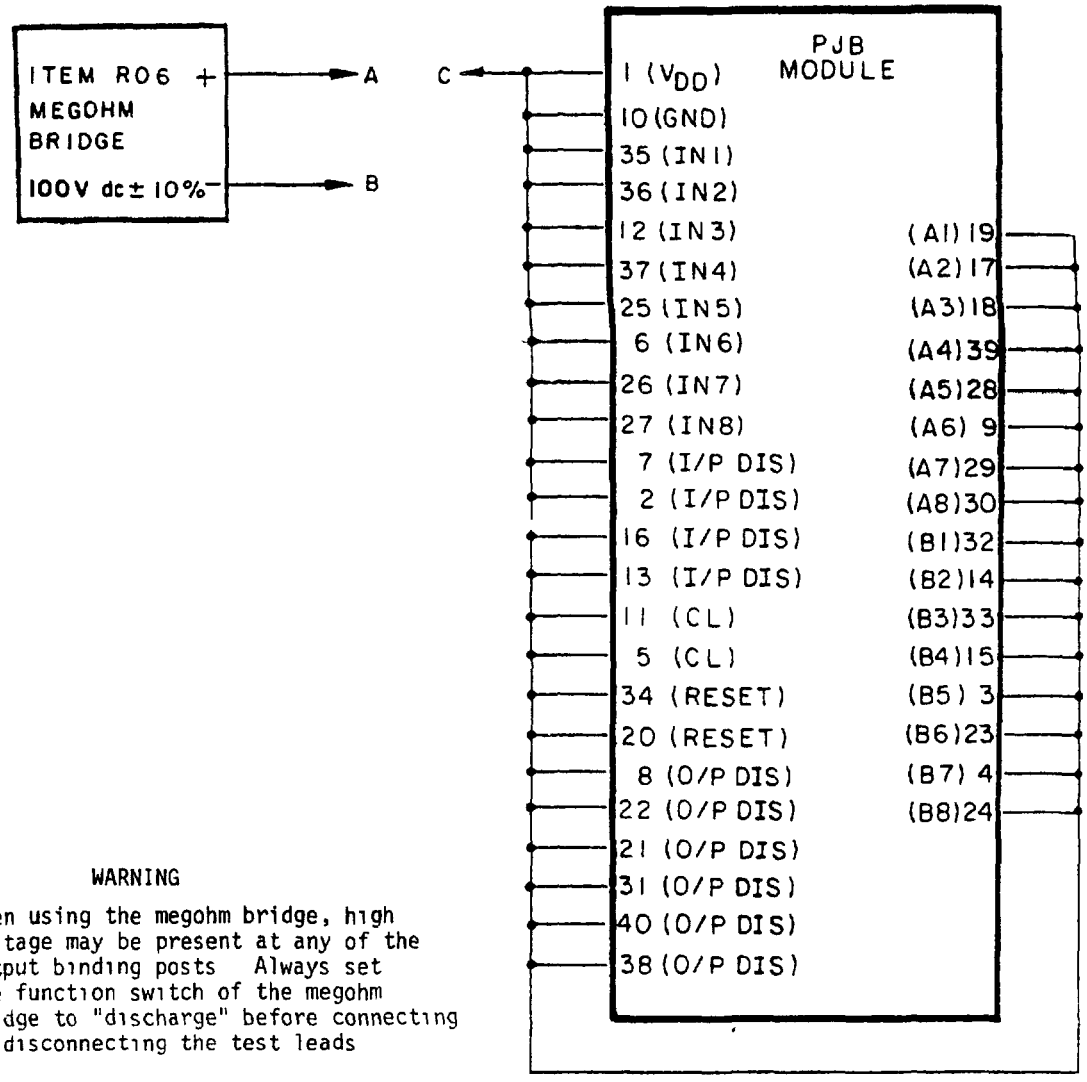
* FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads

NOTE: Penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative lead.

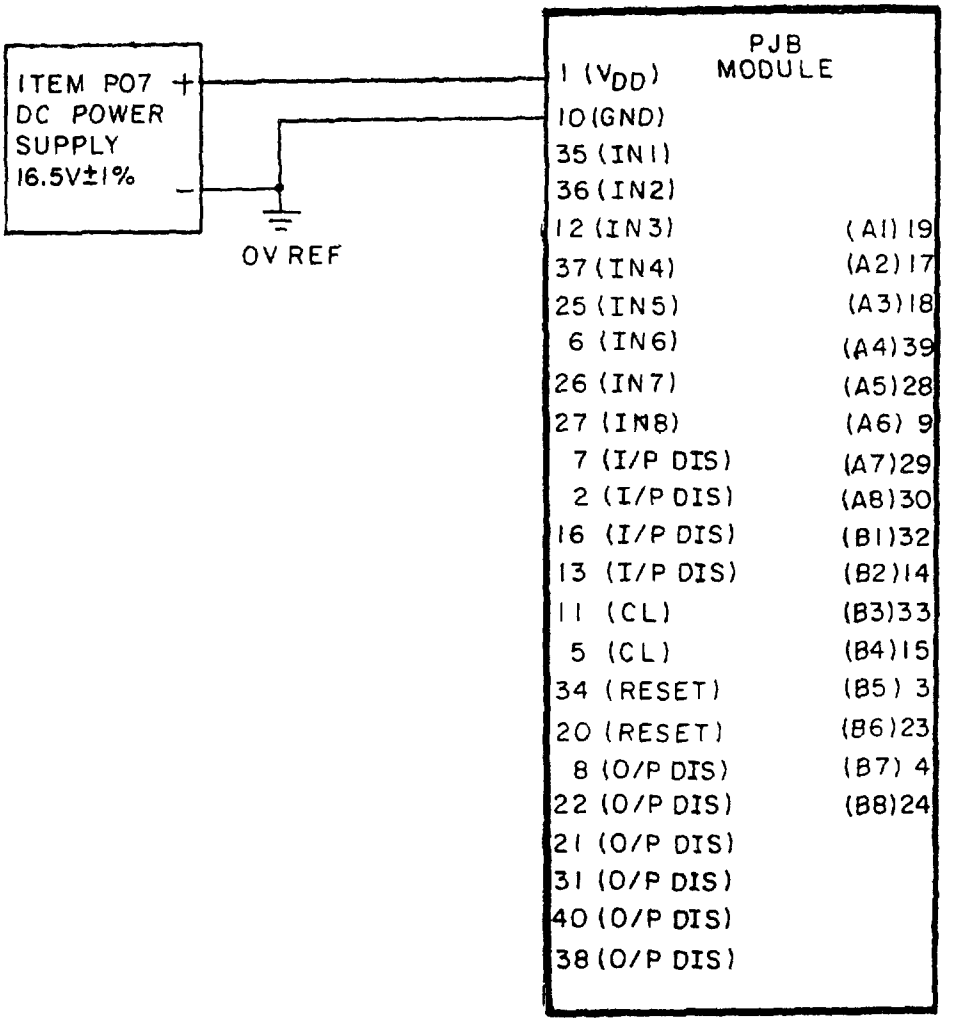
FIGURE 6. Isolation test circuit for table III.

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* TABLE IV. V_{DD} stress test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
B	V_{DD} stress	Power supply voltage for a period of 5.0 ±0.1 seconds	Module must pass all remaining tests			7

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* FIGURE 7. V_{DD} stress test circuit for table IV.

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TABLE V. Input clamp diode voltage (negative) test.

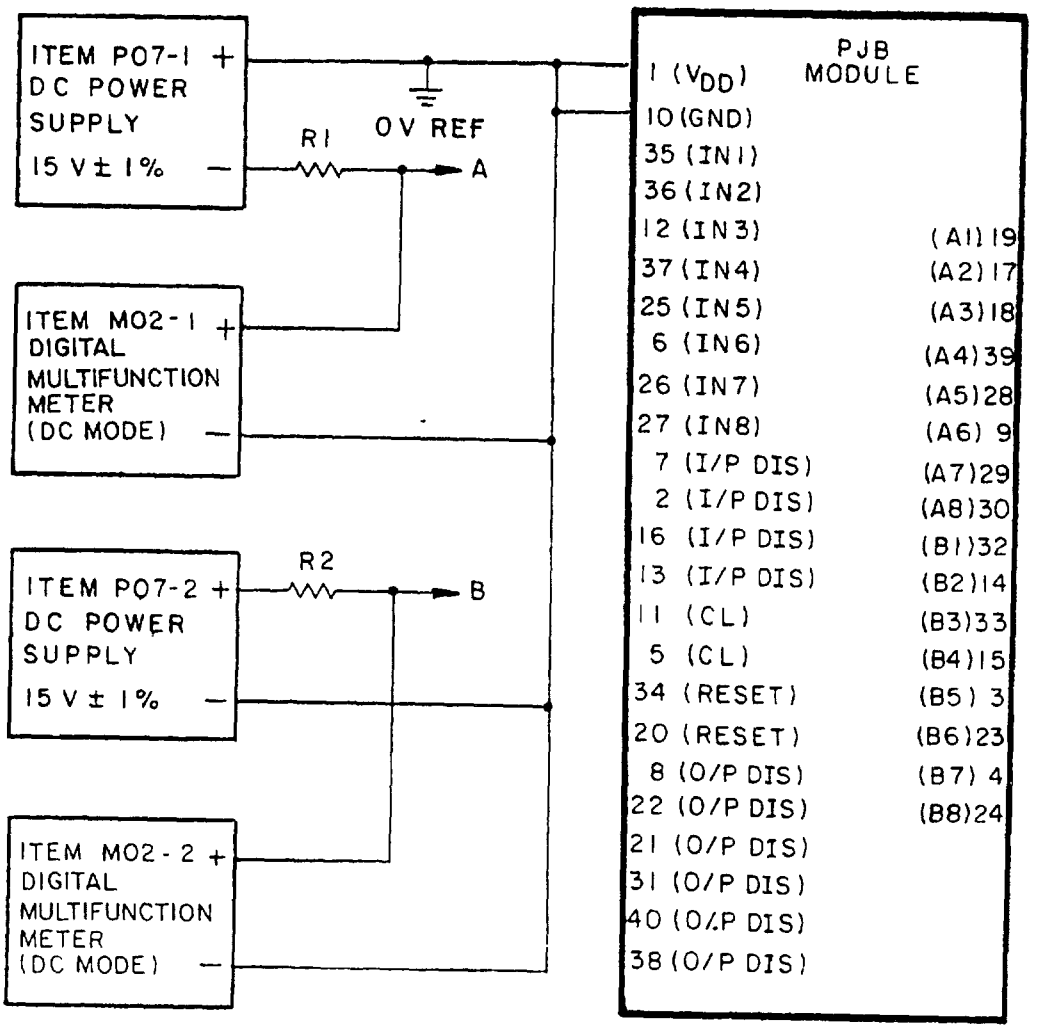
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	IN1 Input clamp diode voltage, positive	35	Open	-0.4	-6.0	V	8
C2	IN2 "	36	"	"	"	"	"
C3	IN3 "	12	"	"	"	"	"
C4	IN4 "	37	"	"	"	"	"
C5	IN5 "	25	"	"	"	"	"
C6	IN6 "	6	"	"	"	"	"
C7	IN7 "	26	"	"	"	"	"
C8	IN8 "	27	"	"	"	"	"
C9	I/P DIS "	7	"	"	"	"	"
C10	I/P DIS "	2	"	"	"	"	"
C11	I/P DIS "	16	"	"	"	"	"
C12	I/P DIS "	13	"	"	"	"	"
C13	CL "	11	"	"	"	"	"
C14	CL "	5	"	"	"	"	"
C15	RESET "	34	"	"	"	"	"
C16	RESET "	20	"	"	"	"	"
C17	O/P DIS "	8	"	"	"	"	"
C18	O/P DIS "	22	"	"	"	"	"
C19	O/P DIS "	21	"	"	"	"	"
C20	O/P DIS "	31	"	"	"	"	"
C21	O/P DIS "	40	"	"	"	"	"
C22	O/P DIS "	38	"	"	"	"	"

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TABLE VI. Input clamp diode voltage (positive) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	IN1	Input clamp diode voltage, positive	Open	35	0.4	1.5	V	8
D2	IN2	"	"	36	"	"	"	"
D3	IN3	"	"	12	"	"	"	"
D4	IN4	"	"	37	"	"	"	"
D5	IN5	"	"	25	"	"	"	"
D6	IN6	"	"	6	"	"	"	"
D7	IN7	"	"	26	"	"	"	"
D8	IN8	"	"	27	"	"	"	"
D9	I/P DIS	"	"	7	"	"	"	"
D10	I/P DIS	"	"	2	"	"	"	"
D11	I/P DIS	"	"	16	"	"	"	"
D12	I/P DIS	"	"	13	"	"	"	"
D13	CL	"	"	11	"	"	"	"
D14	CL	"	"	5	"	"	"	"
D15	RESET	"	"	34	"	"	"	"
D16	RESET	"	"	20	"	"	"	"
D17	O/P DIS	"	"	8	"	"	"	"
D18	O/P DIS	"	"	22	"	"	"	"
D19	O/P DIS	"	"	21	"	"	"	"
D20	O/P DIS	"	"	31	"	"	"	"
D21	O/P DIS	"	"	40	"	"	"	"
D22	O/P DIS	"	"	38	"	"	"	"

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NOTES
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.

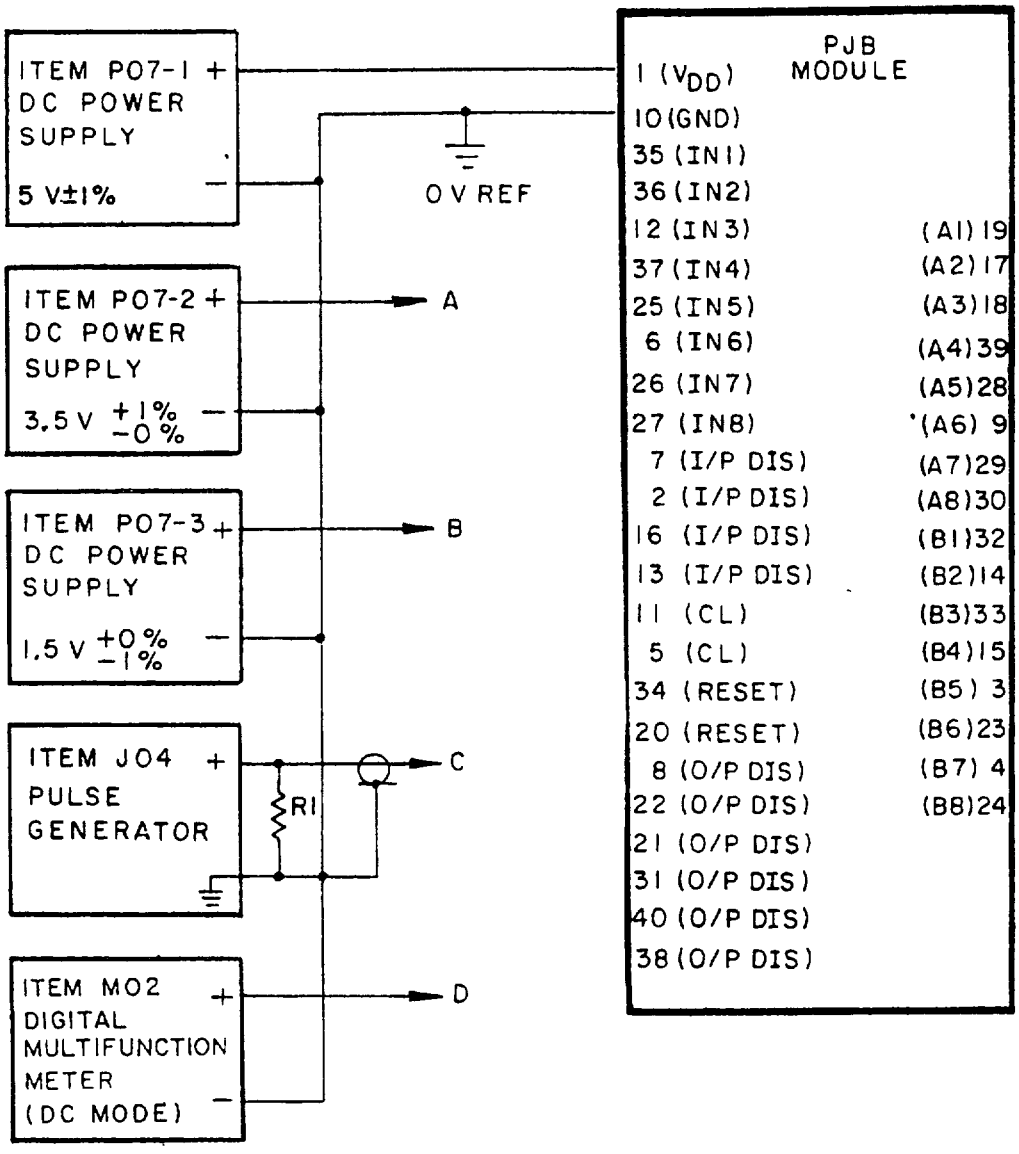
* FIGURE 8. Input clamp diode voltage (negative and positive) test circuit for tables V and VI.

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TABLE VII. Functional test.

Test condition	Parameter	Connect			Output pins								Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	19 32	17 14	18 33	39 15	28 3	9 23	29 4	30 24	Min	Max		
E1	Functional	11,5,35,36,12, 37,25,6,26,27, 34,20	8,22,21,31,40 38,7,2,16,13	Open	L	L	L	L	L	L	L	L	See figure 9, Note 2			9,10
E2	"	35,36,12,37,25, 6,26,27	8,22,21,31,40, 38,7,2,16,13, 34,20,11,5	Open	L	L	L	L	L	L	L	L	"			"
E3	"	35,36,12,37,25, 6,26,27,7,2,16, 13	34,20,8,22,21, 31,40,38	5,11	L	L	L	L	L	L	L	L	"			"
E4	"	35,36,12,37,25, 6,26,27	7,2,16,13,34, 20,8,22,21,31, 40,38	"	H	H	H	H	H	H	H	H	"			"
E5	"	35,36,12,37,25, 6,26,27,7,2,16, 13	34,20,8,22,21, 31,40,38	"	H	H	H	H	H	H	H	H	"			"
E6	"	Open	34,35,36,37, 38,25,26,27, 6,7,8,12,13,2, 20,21,22,40, 31,16	"	L	L	L	L	L	L	L	L	"			"
E7	"	7,2,16,13	35,36,12,37, 25,6,26,27,34, 20,8,22,21,31, 40,38	"	L	L	L	L	L	L	L	L	"			"

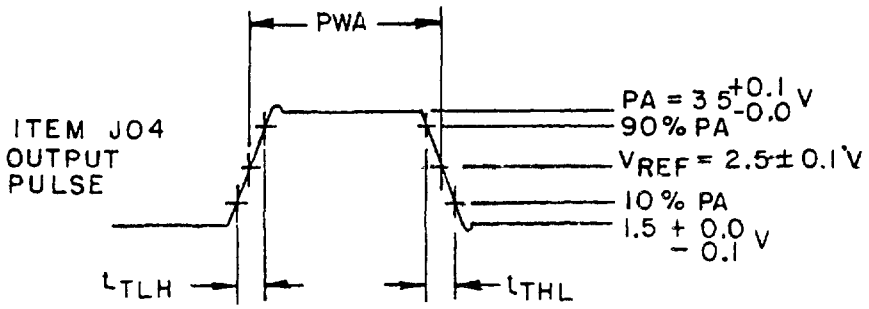
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. Verify that the output high level is ≥ 4.6 V and the low level is ≤ 0.4 V using item MO2.
3. When point C is connected, pulse the module one time.

* FIGURE 9. Functional test circuit for table VII.

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NOTE Input pulse parameters.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
Single pulse mode.

FIGURE 10. Functional test circuit waveform for tables VII, X, and XI.

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TABLE VIII Output voltage (low-level, I_{OL} = 0) test.

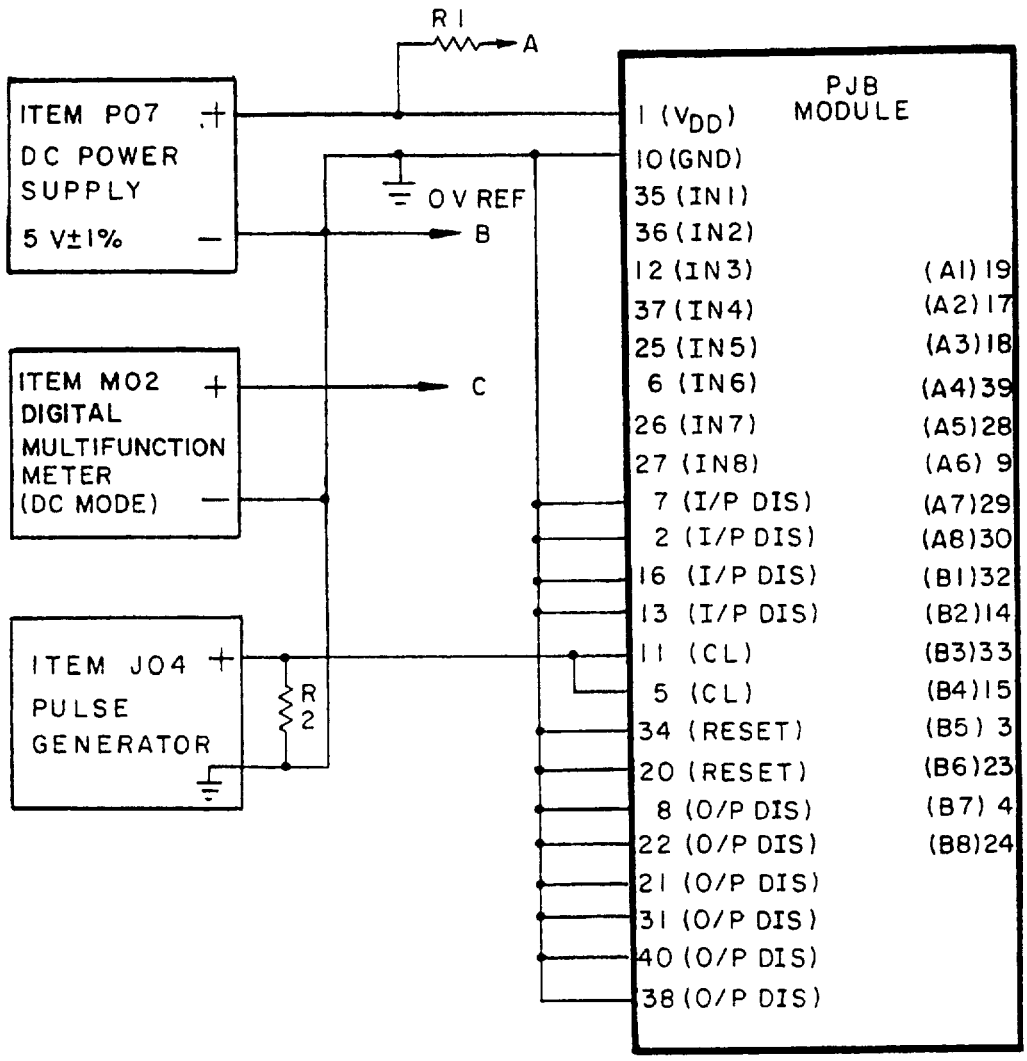
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	A1	Output voltage, low level (I_{OL} = 0)	Open	35,36,12, 37,25,6, 26,27	19	0.00	0.05	V	11,12
F2	A2	"	"	"	17	"	"	"	"
F3	A3	"	"	"	18	"	"	"	"
F4	A4	"	"	"	39	"	"	"	"
F5	A5	"	"	"	28	"	"	"	"
F6	A6	"	"	"	9	"	"	"	"
F7	A7	"	"	"	29	"	"	"	"
F8	A8	"	"	"	30	"	"	"	"
F9	B1	"	"	"	32	"	"	"	"
F10	B2	"	"	"	14	"	"	"	"
F11	B3	"	"	"	33	"	"	"	"
F12	B4	"	"	"	15	"	"	"	"
F13	B5	"	"	"	3	"	"	"	"
F14	B6	"	"	"	23	"	"	"	"
F15	B7	"	"	"	4	"	"	"	"
F16	B8	"	"	"	24	"	"	"	"

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TABLE IX. Output voltage (high-level, I_{OH} = 0) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	A1	Output voltage, high level (I_{OH} = 0)	35,36,12,37, 25,6,26,27	Open	19	4.95	5.05	V	11,12
G2	A2	"	"	"	17	"	"	"	"
G3	A3	"	"	"	18	"	"	"	"
G4	A4	"	"	"	39	"	"	"	"
G5	A5	"	"	"	28	"	"	"	"
G6	A6	"	"	"	9	"	"	"	"
G7	A7	"	"	"	29	"	"	"	"
G8	A8	"	"	"	30	"	"	"	"
G9	B1	"	"	"	32	"	"	"	"
G10	B2	"	"	"	14	"	"	"	"
G11	B3	"	"	"	33	"	"	"	"
G12	B4	"	"	"	15	"	"	"	"
G13	B5	"	"	"	3	"	"	"	"
G14	B6	"	"	"	23	"	"	"	"
G15	B7	"	"	"	4	"	"	"	"
G16	B8	"	"	"	24	"	"	"	"

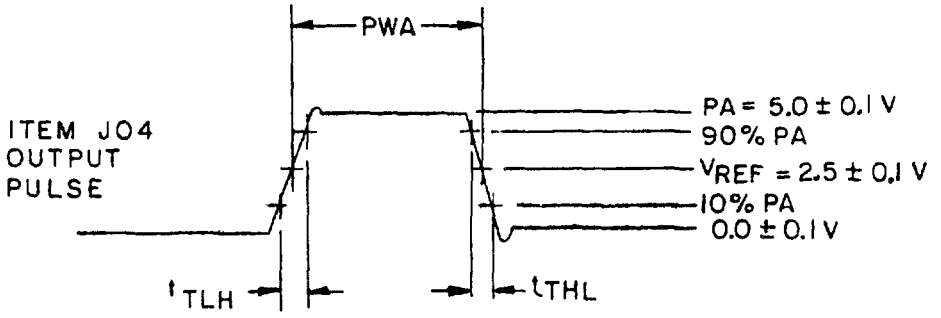
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1/2 W.
3. Clock the module one time after all connections have been made.

* FIGURE 11. Output voltage (low-level, I_{OL} = 0 and high-level, I_{OH} = 0) test circuit for tables VIII and IX.

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NOTE Input pulse parameters
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
Single pulse mode.

* FIGURE 12. Output voltage (low-level, I_{OL} = 0 and high-level, I_{OH} = 0) test circuit waveforms for tables VIII and IX.

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TABLE X. Output voltage (low-level, V_{DD} = 5 V) test.

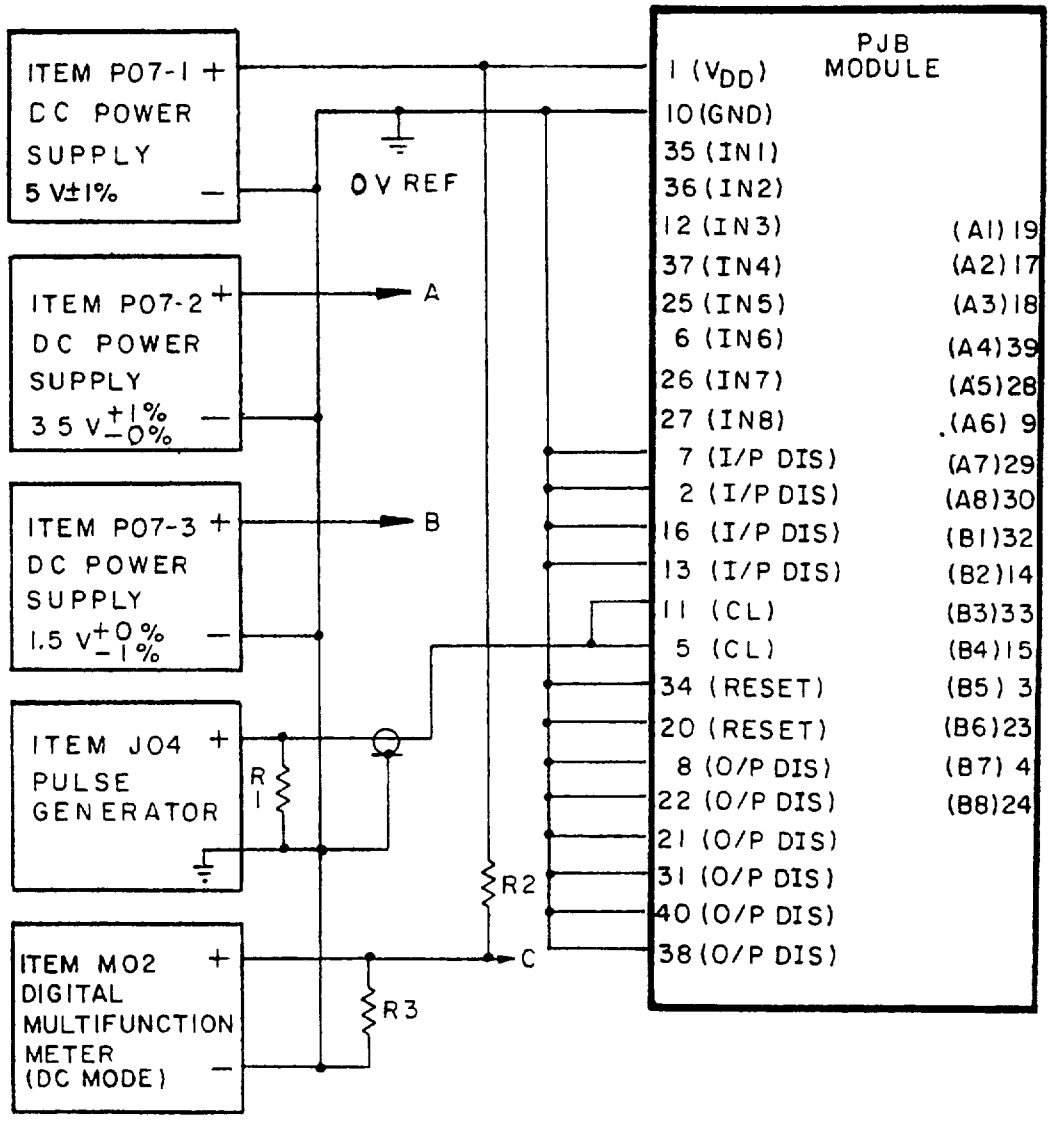
Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	A1	Output voltage, low level (V_{DD} = 5 V)	Open	35,36,12, 37,25,6, 26,27	19	0.0	0.4	V	10,13
H2	A2	"	"	"	17	"	"	"	"
H3	A3	"	"	"	18	"	"	"	"
H4	A4	"	"	"	39	"	"	"	"
H5	A5	"	"	"	28	"	"	"	"
H6	A6	"	"	"	9	"	"	"	"
H7	A7	"	"	"	29	"	"	"	"
H8	A8	"	"	"	30	"	"	"	"
H9	B1	"	"	"	32	"	"	"	"
H10	B2	"	"	"	14	"	"	"	"
H11	B3	"	"	"	33	"	"	"	"
H12	B4	"	"	"	15	"	"	"	"
H13	B5	"	"	"	3	"	"	"	"
H14	B6	"	"	"	23	"	"	"	"
H15	B7	"	"	"	4	"	"	"	"
H16	B8	"	"	"	24	"	"	"	"

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TABLE XI. Output voltage (high-level, V_{DD} = 5 V) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	A1	Output voltage, high level (V_{DD} = 5 V)	35,36,12,37, 25,6,26,27	Open	19	4.6	5.0	V	10,13
J2	A2	"	"	"	17	"	"	"	"
J3	A3	"	"	"	18	"	"	"	"
J4	A4	"	"	"	39	"	"	"	"
J5	A5	"	"	"	28	"	"	"	"
J6	A6	"	"	"	9	"	"	"	"
J7	A7	"	"	"	29	"	"	"	"
J8	A8	"	"	"	30	"	"	"	"
J9	B1	"	"	"	32	"	"	"	"
J10	B2	"	"	"	14	"	"	"	"
J11	B3	"	"	"	33	"	"	"	"
J12	B4	"	"	"	15	"	"	"	"
J13	B5	"	"	"	3	"	"	"	"
J14	B6	"	"	"	23	"	"	"	"
J15	B7	"	"	"	4	"	"	"	"
J16	B8	"	"	"	24	"	"	"	"

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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. R2 (0-55°C) = 7.18 kΩ ±5%, 1/8 W.
   (00°-25°C) = 9.02 kΩ ±5%, 1/8 W.
   (060°C) = 10.45 kΩ ±1%, 1/8 W.
   (085°C) = 12.78 kΩ ±1%, 1/8 W.
3. R3 (0-55°C) = 18.4 kΩ ±1%, 1/8 W.
   (00°-25°C) = 23.0 kΩ ±1%, 1/8 W.
   (060°C) = 27.06 kΩ ±1%, 1/8 W.
   (085°C) = 32.86 kΩ ±1%, 1/8 W.
4. After all connections have been made, clock the module one time per figure 10.

* FIGURE 13. Output voltage (low-level and high-level, V_{DD} = 5 V) test circuit for tables X and XI.

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TABLE XII. Output voltage (low-level, V_{DD} = 15 V) test.

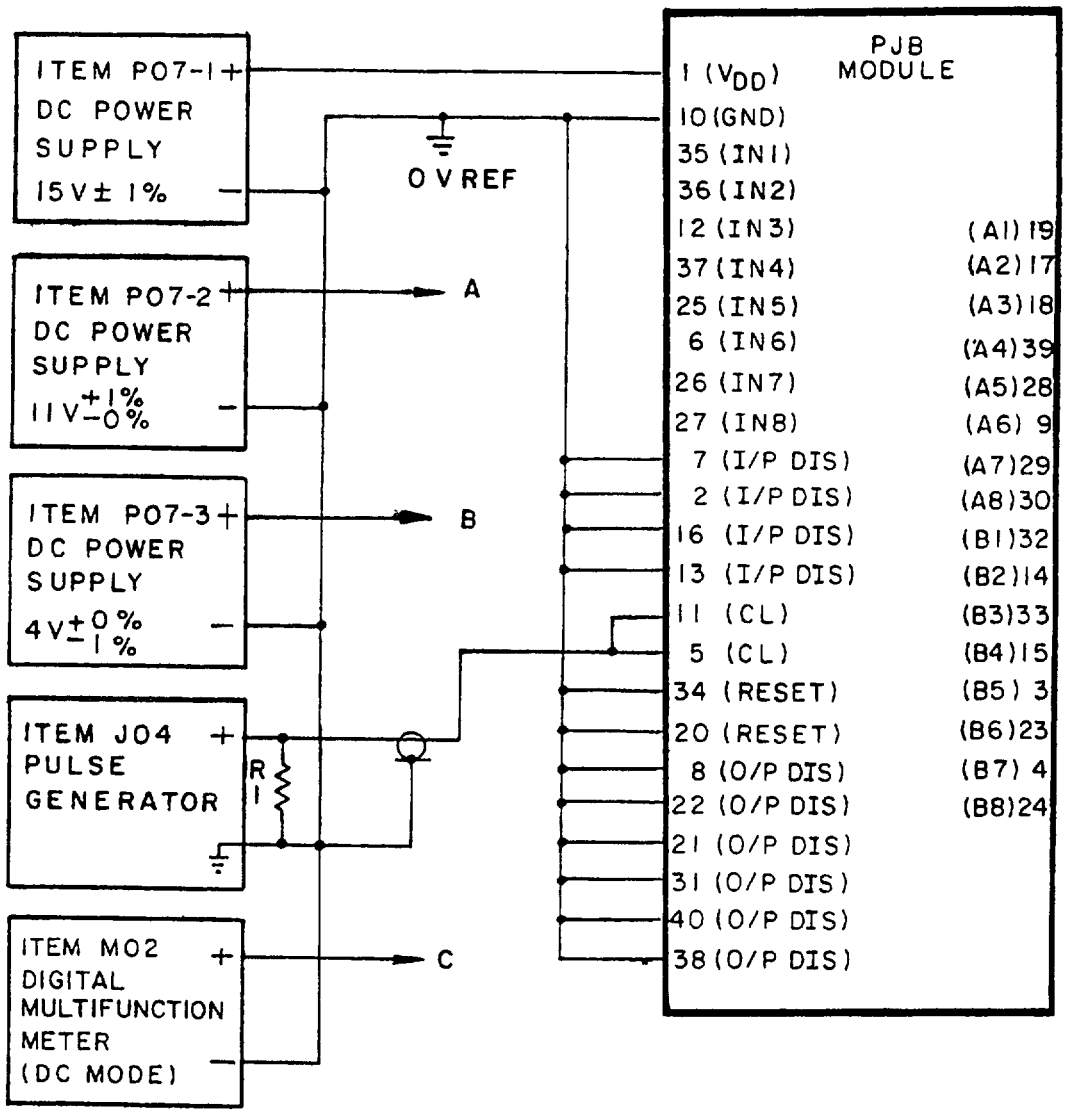
Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	A1	Output voltage, low level (V_{DD} = 15 V)	Open	35,36,12, 37,25,6, 26,27	19	0.0	1.5	V	14,15
K2	A2	"	"	"	17	"	"	"	"
K3	A3	"	"	"	18	"	"	"	"
K4	A4	"	"	"	39	"	"	"	"
K5	A5	"	"	"	28	"	"	"	"
K6	A6	"	"	"	9	"	"	"	"
K7	A7	"	"	"	29	"	"	"	"
K8	A8	"	"	"	30	"	"	"	"
K9	B1	"	"	"	32	"	"	"	"
K10	B2	"	"	"	14	"	"	"	"
K11	B3	"	"	"	33	"	"	"	"
K12	B4	"	"	"	15	"	"	"	"
K13	B5	"	"	"	3	"	"	"	"
K14	B6	"	"	"	23	"	"	"	"
K15	B7	"	"	"	4	"	"	"	"
K16	B8	"	"	"	24	"	"	"	"

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TABLE XIII. Output voltage (high-level, V_{DD} = 15 V) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L1	A1	Output voltage, high level (V_{DD} = 15 V)	35,36,12,37, 25,6,26,27	Open	19	13.5	15.0	V	14,15
L2	A2	"	"	"	17	"	"	"	"
L3	A3	"	"	"	18	"	"	"	"
L4	A4	"	"	"	39	"	"	"	"
L5	A5	"	"	"	28	"	"	"	"
L6	A6	"	"	"	9	"	"	"	"
L7	A7	"	"	"	29	"	"	"	"
L8	A8	"	"	"	30	"	"	"	"
L9	B1	"	"	"	32	"	"	"	"
L10	B2	"	"	"	14	"	"	"	"
L11	B3	"	"	"	33	"	"	"	"
L12	B4	"	"	"	15	"	"	"	"
L13	B5	"	"	"	3	"	"	"	"
L14	B6	"	"	"	23	"	"	"	"
L15	B7	"	"	"	4	"	"	"	"
L16	B8	"	"	"	24	"	"	"	"

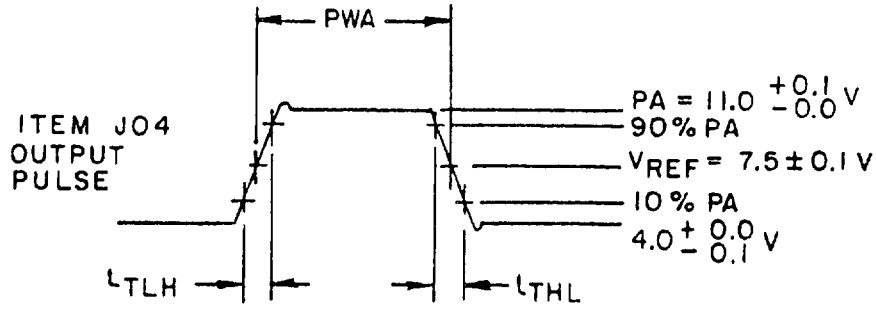
Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:42Z
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NOTES:
1. R1 = 57Ω ±5%, 5 W.
2. After all connections have been made, clock the module one time per figure 15.

* FIGURE 14. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables XII and XIII.

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NOTE: Input pulse parameters:
PWA = 500 ±50 ns
t_{TLH}, t_{THL} = 10 ±2 ns.
Single pulse mode.

* FIGURE 15. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables XII and XIII.

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TABLE XIV. Input current (low-level) test.

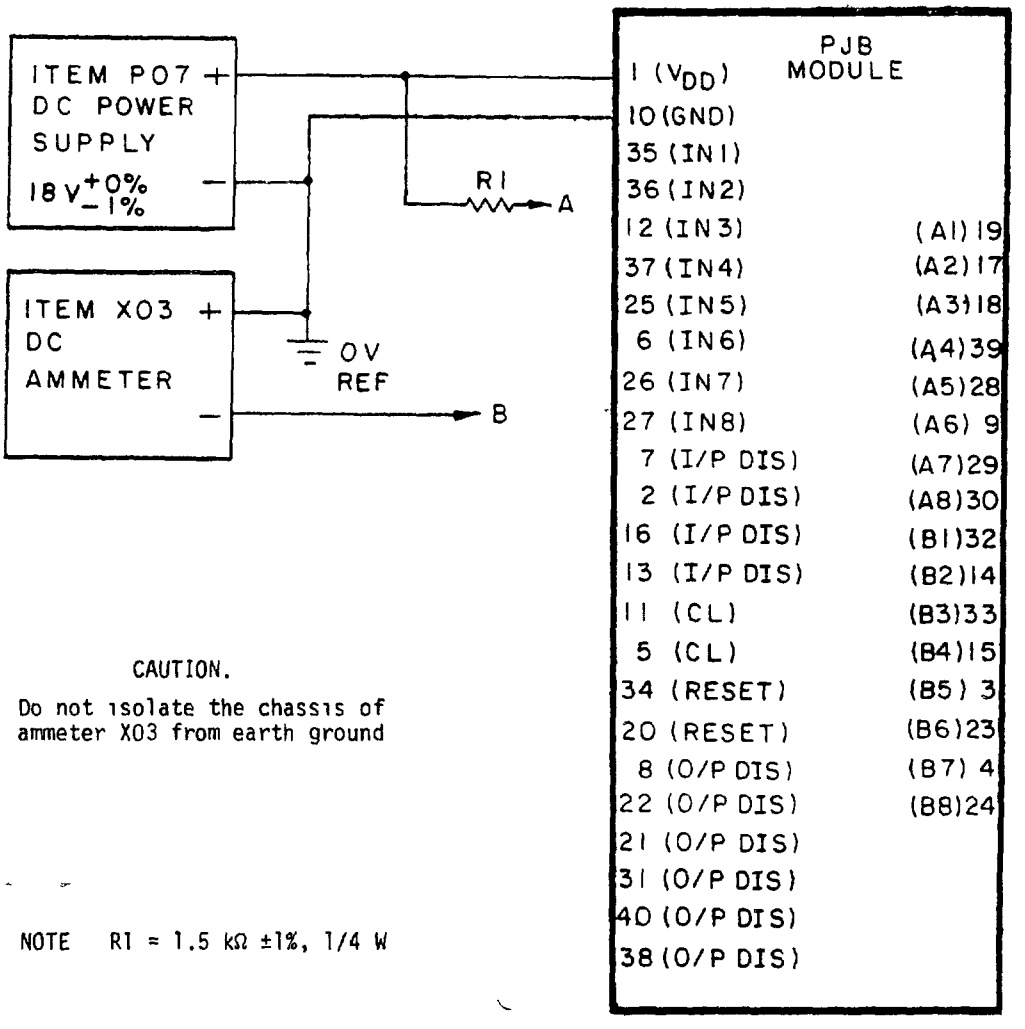
Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	IN1	Input current, low level	34,20,8,22,21,31,40,38, 36,12,37,25,6,26,27,7, 2,16,13,11,5	35	0	-200	nA	16
M2	IN2	"	34,20,8,22,21,31,40,38, 35,12,37,25,6,26,27,7, 2,16,13,11,5	36	"	"	"	"
M3	IN3	"	34,20,8,22,21,31,40,38, 35,36,37,25,6,26,27,7 ,2,16,13,11,5	12	"	"	"	"
M4	IN4	"	34,20,8,22,21,31,40,38, 35,36,12,25,6,26,27,7, 2,16,13,11,5	37	"	"	"	"
M5	IN5	"	34,20,8,22,21,31,40,38, 35,36,12,37,6,26,27,7, 2,16,13,11,5	25	"	"	"	"
M6	IN6	"	34,20,8,22,21,31,40,38, 35,36,12,37,25,26,27, 7,2,16,13,11,5	6	"	"	"	"
M7	IN7	"	34,20,8,22,21,31,40,38, 35,36,12,37,25,6,27, 7,2,16,13,11,5	26	"	"	"	"
M8	IN8	"	34,20,8,22,21,31,40,38, 35,36,12,37,25,6,26, 7,2,16,13,11,5	27	"	"	"	"
M9	I/P DIS	"	34,20,8,22,21,31,40, 38,35,36,12,37,25,6, 26,27,2,16,13,11,5	7	"	"	"	"
M10	I/P DIS	"	34,20,8,22,21,31,40,38, 35,36,12,37,25,6,26, 27,7,16,13,11,5	2	"	"	"	"
M11	I/P DIS	"	34,20,8,22,21,31,40, 38,35,36,12,37,25,6, 26,27,7,2,13,11,5	16	"	"	"	"
M12	I/P DIS	"	34,20,8,22,21,31,40,38, 35,36,12,37,25,6,26, 27,7,2,16,11,5	13	"	"	"	"

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TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M13	CL	Input current, low level	34,20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,5	11	0	-200	nA	16
M14	CL	"	34,20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11	5	"	"	"	"
M15	RESET	"	20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	34	"	"	"	"
M16	RESET	"	34,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	20	"	"	"	"
M17	O/P DIS	"	34,20,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	8	"	"	"	"
M18	O/P DIS	"	34,20,8,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	22	"	"	"	"
M19	O/P DIS	"	34,20,8,22,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	21	"	"	"	"
M20	O/P DIS	"	34,20,8,22,21,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	31	"	"	"	"
M21	O/P DIS	"	34,20,8,22,21,31,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	40	"	"	"	"
M22	O/P DIS	"	34,20,8,22,21,31,40,35,36,12,37,25,6,26,27,7,2,16,13,11,5	38	"	"	"	"

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CAUTION.

Do not isolate the chassis of ammeter X03 from earth ground

NOTE R1 = 1.5 kΩ ±1%, 1/4 W

FIGURE 16. Input current (low-level) test circuit for table XIV.

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TABLE XV. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	IN1	Input current, high level	34,20,8,22,21,31,40,38,36, 12,37,25,6,26,27,7,2,16, 13,11,5	35	0	200	nA	17
N2	IN2	"	34,20,8,22,21,31,40,38,35, 12,37,25,6,26,27,7,2,16, 13,11,5	36	"	"	"	"
N3	IN3	"	34,20,8,22,21,31,40,38,35, 36,37,25,6,26,27,7,2,16, 13,11,5	12	"	"	"	"
N4	IN4	"	34,20,8,22,21,31,40,38,35, 36,12,25,6,26,27,7,2,16, 13,11,5	37	"	"	"	"
N5	IN5	"	34,20,8,22,21,31,40,38,35, 36,12,37,6,26,27,7,2,16, 13,11,5	25	"	"	"	"
N6	IN6	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,26,27,7,2,16, 13,11,5	6	"	"	"	"
N7	IN7	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,6,27,7,2,16, 13,11,5	26	"	"	"	"
N8	IN8	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,6,26,7,2,16, 13,11,5	27	"	"	"	"
N9	I/P DIS	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,6,26,27,2,16, 13,11,5	7	"	"	"	"
N10	I/P DIS	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,6,26,27,7,16, 13,11,5	2	"	"	"	"
N11	I/P DIS	"	34,20,8,22,21,31,40,38,35, 36,12,37,25,6,26,27,7,2, 13,11,5	16	"	"	"	"

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TABLE XV. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N12	I/P DIS	Input current, high level	34,20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,11,5	13	0	200	nA	17
N13	CL	"	34,20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,5	11	"	"	"	"
N14	CL	"	34,20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11	5	"	"	"	"
N15	RESET	"	20,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	34	"	"	"	"
N16	RESET	"	34,8,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	20	"	"	"	"
N17	O/P DIS	"	34,20,22,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	8	"	"	"	"
N18	O/P DIS	"	34,20,8,21,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	22	"	"	"	"
N19	O/P DIS	"	34,20,8,22,31,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	21	"	"	"	"
N20	O/P DIS	"	34,20,8,22,21,40,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	31	"	"	"	"
N21	O/P DIS	"	34,20,8,22,21,31,38,35,36,12,37,25,6,26,27,7,2,16,13,11,5	40	"	"	"	"
N22	O/P DIS	"	34,20,8,22,21,31,40,35,36,12,37,25,6,26,27,7,2,16,13,11,5	38	"	"	"	"

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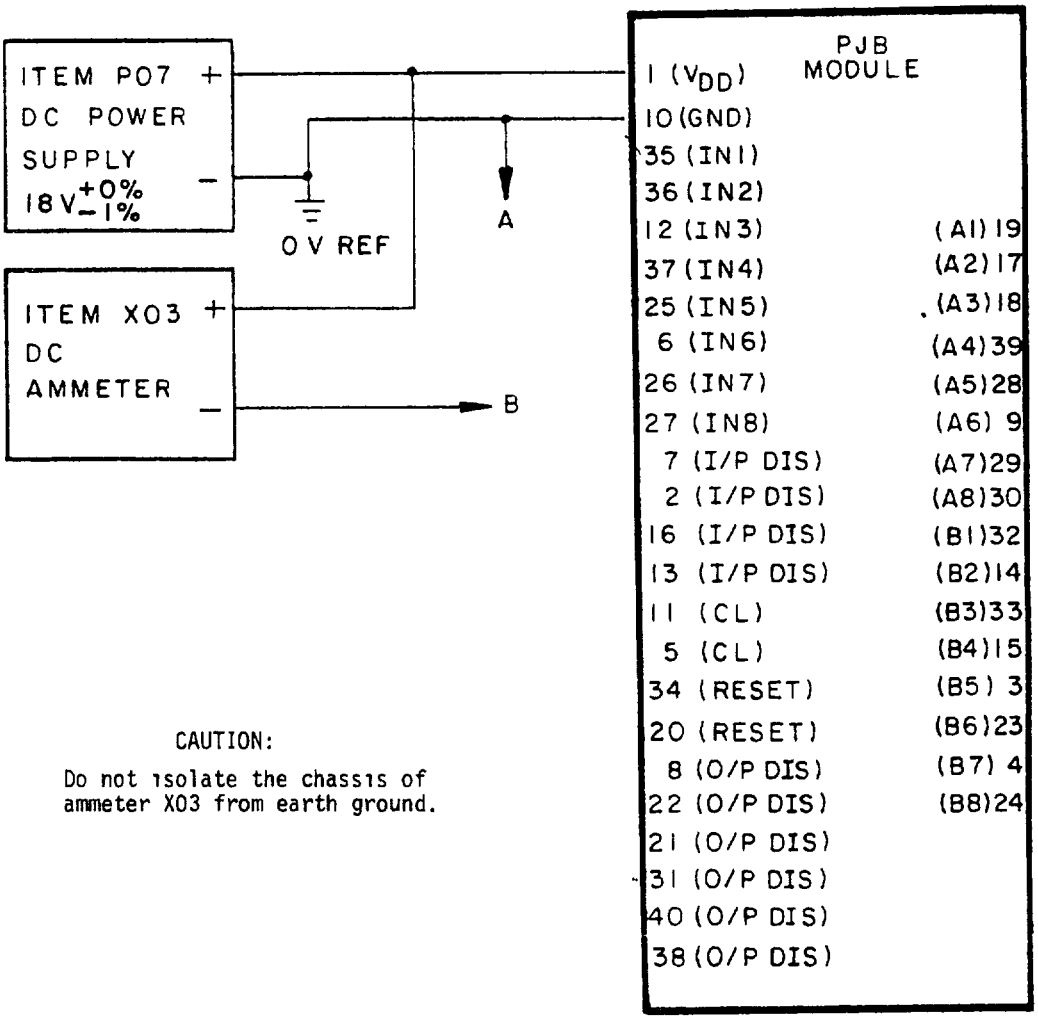


FIGURE 17. Input current (high-level) test circuit for table XV.

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TABLE XVI. Output leakage current (low-level, I_{OZL}) test.

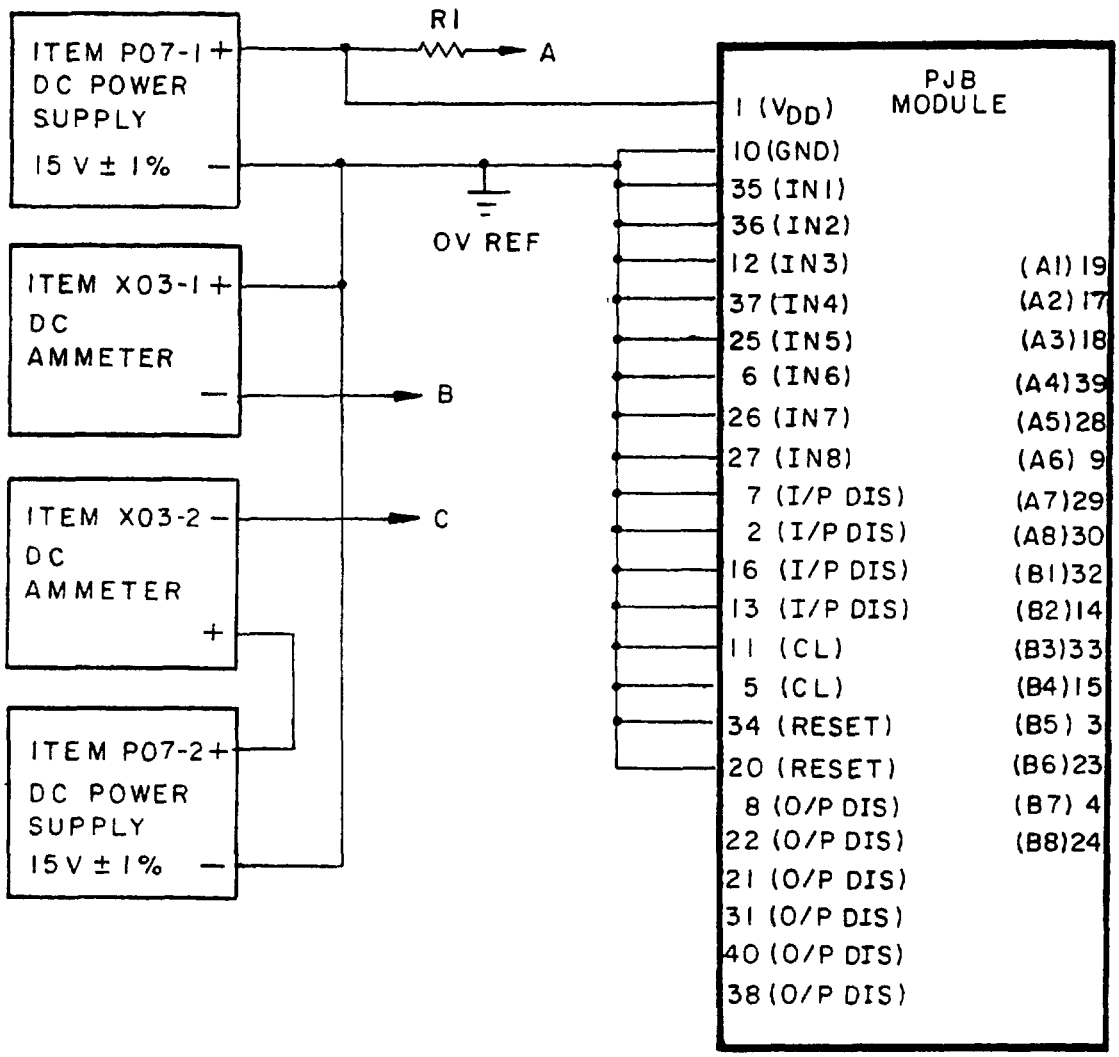
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	A1	Output leakage current, low level (I_{OZL})	8,22,21,31, 40,38	19	Open	0.0	-0.4	µA	18
P2	A2	"	"	17	"	"	"	"	"
P3	A3	"	"	18	"	"	"	"	"
P4	A4	"	"	39	"	"	"	"	"
P5	A5	"	"	28	"	"	"	"	"
P6	A6	"	"	9	"	"	"	"	"
P7	A7	"	"	29	"	"	"	"	"
P8	A8	"	"	30	"	"	"	"	"
P9	B1	"	"	32	"	"	"	"	"
P10	B2	"	"	14	"	"	"	"	"
P11	B3	"	"	33	"	"	"	"	"
P12	B4	"	"	15	"	"	"	"	"
P13	B5	"	"	3	"	"	"	"	"
P14	B6	"	"	23	"	"	"	"	"
P15	B7	"	"	4	"	"	"	"	"
P16	B8	"	"	24	"	"	"	"	"

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TABLE XVII. Output leakage current (high-level, I_{OZH}) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	A1	Output leakage current, high level (I_{OZH})	8,22,21,31, 40,38	Open	19	0.0	0.4	µA	18
Q2	A2	"	"	"	17	"	"	"	"
Q3	A3	"	"	"	18	"	"	"	"
Q4	A4	"	"	"	39	"	"	"	"
Q5	A5	"	"	"	28	"	"	"	"
Q6	A6	"	"	"	9	"	"	"	"
Q7	A7	"	"	"	29	"	"	"	"
Q8	A8	"	"	"	30	"	"	"	"
Q9	B1	"	"	"	32	"	"	"	"
Q10	B2	"	"	"	14	"	"	"	"
Q11	B3	"	"	"	33	"	"	"	"
Q12	B4	"	"	"	15	"	"	"	"
Q13	B5	"	"	"	3	"	"	"	"
Q14	B6	"	"	"	23	"	"	"	"
Q15	B7	"	"	"	4	"	"	"	"
Q16	B8	"	"	"	24	"	"	"	"

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CAUTION:
Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/8 W.

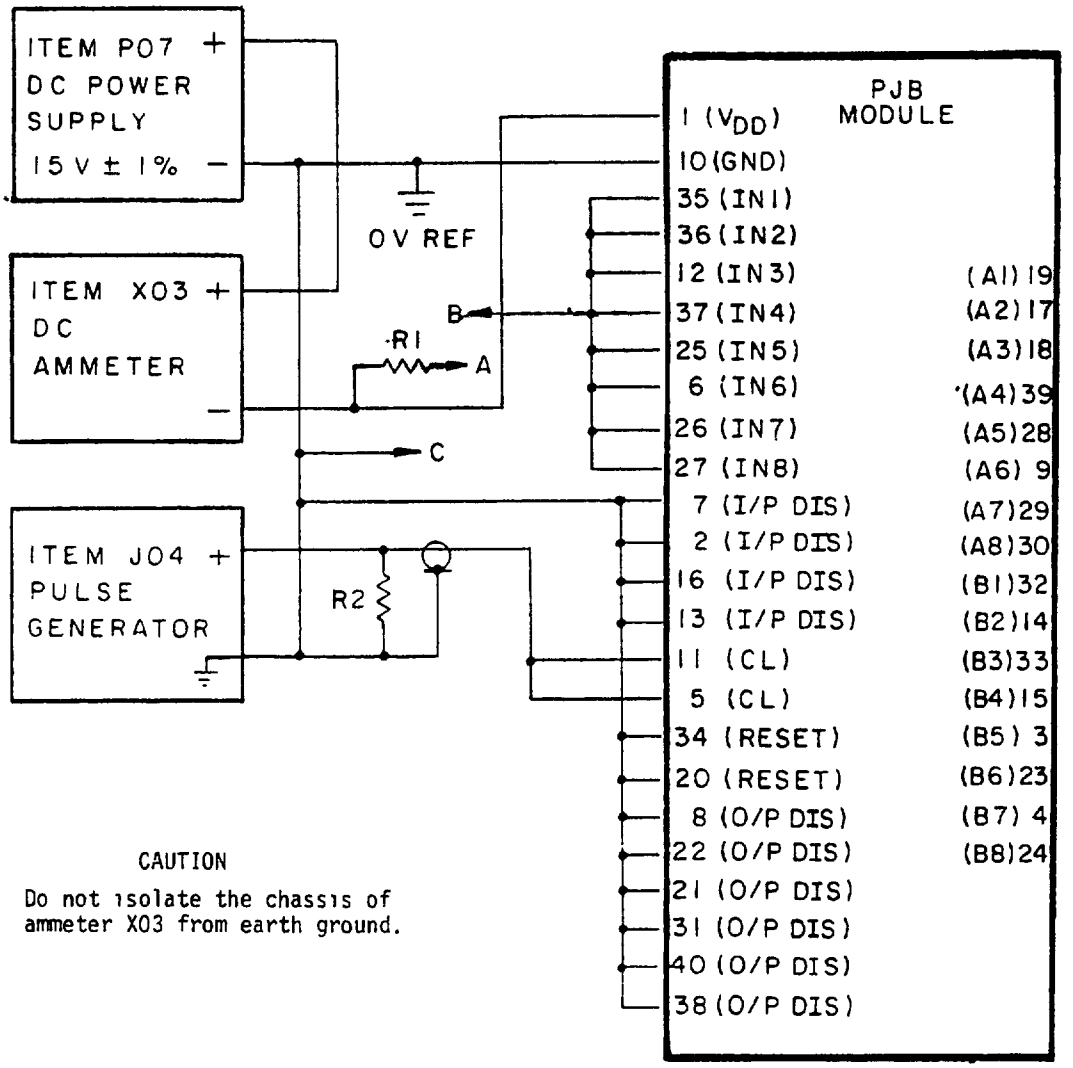
FIGURE 18. Output leakage (low-level, I_{OZL} and high-level, I_{OZH}) test circuit for tables XVI and XVII.

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TABLE XVIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Power supply current, low level	Open	C	B	0	80	µA	19,20
R2	Power supply current, high level	B	A	Open	0	80	µA	19,20

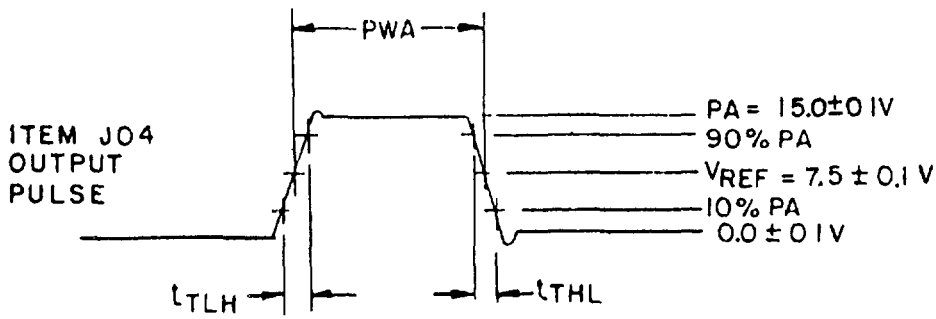
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 5 W.
3. After all connections have been made, clock the module one time per figure 20.

FIGURE 19. Power supply current (low-level and high-level) test circuit for table XVIII.

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NOTE:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

   Single pulse mode.

FIGURE 20. Power supply current (low-level and high-level) test circuit waveforms for table XVIII.

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TABLE XIX. Transition time (t_{TLH}, V_{DD} = 5 V) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	A1	Transition time (t_{TLH}), V_{DD} = 5 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	36	360	ns	21,22
S2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
S3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
S4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
S5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
S6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
S7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
S8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
S9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
S10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
S11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
S12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
S13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
S14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
S15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
S16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

TABLE XIX. Transition time (t_{TLH}, V_{DD} = 5 V) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
S17	A1	Transition time (t_{TLH}), V_{DD} = 5 V, C_L = 150 pF	35	19	26,27,36,12, 37,25,6	84	840	ns	21,22
S18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
S19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
S20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
S21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
S22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
S23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
S24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
S25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
S26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
S27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
S28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
S29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
S30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
S31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
S32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
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TABLE XX. Transition time (t_{THL}, V_{DD} = 5 V) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	A1	Transition time (t_{THL}), V_{DD} = 5 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	20	200	ns	21,22
T2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
T3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
T4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
T5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
T6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
T7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
T8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
T9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
T10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
T11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
T12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
T13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
T14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
T15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
T16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XX. Transition time (t_{THL}, V_{DD} = 5 V) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T17	A1	Transition time (t_{THL}), V_{DD} = 5 V, C_L = 150 pF	35	19	26,27,36,12, 37,25,6	44	440	ns	21,22
T18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
T19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
T20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
T21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
T22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
T23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
T24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
T25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
T26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
T27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
T28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
T29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
T30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
T31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
T32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXI. Transition time (t_{TLH}, V_{DD} = 10 V) test.

Test condition	Parameter		Connect			Initial limits 25C °±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U1	A1	Transition time (t_{TLH}), V_{DD} = 10 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	18	180	ns	21,22
U2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
U3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
U4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
U5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
U6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
U7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
U8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
U9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
U10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
U11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
U12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
U13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
U14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
U15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
U16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXII. Transition time (t_{THL}, V_{DD} = 10 V) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
V1	A1	Transition time (t_{THL}), V_{DD} = 10 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	10	100	ns	21,22
V2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
V3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
V4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
V5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
V6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
V7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
V8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
V9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
V10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
V11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
V12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
V13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
V14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
V15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
V16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXIII. Transition time (t_{TLH}, V_{DD} = 15 V) test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
W1	A1	Transition time (t_{TLH}), V_{DD} = 15 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	13	130	ns	21,22
W2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
W3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
W4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
W5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
W6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
W7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
W8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
W9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
W10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
W11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
W12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
W13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
W14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
W15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
W16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXIV. Transition time (t_{THL}, V_{DD} = 15 V) test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
X1	A1	Transition time (t_{THL}), V_{DD} = 15 V, C_L = 50 pF	35	19	26,27,36,12, 37,25,6	8	80	ns	21,22
X2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
X3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
X4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
X5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
X6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
X7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
X8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
X9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
X10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
X11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
X12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
X13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
X14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
X15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
X16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXV. Propagation delay time (CL to output, V_{DD} = 5 V, t_{PLH} and t_{PHL}) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Y1	A1	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 50 pF, t_{PLH}	35	19	26,27,36,12, 37,25,6	60	600	ns	21,22
Y2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
Y3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
Y4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
Y5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
Y6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
Y7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
Y8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
Y9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
Y10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
Y11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
Y12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
Y13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
Y14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
Y15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
Y16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXV. Propagation delay time (CL to output, V_{DD} = 5 V, t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Y17	A1	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 50 pF, t_{PHL}	35	19	26,27,36,12, 37,25,6	60	600	ns	21,22
Y18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
Y19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
Y20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
Y21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
Y22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
Y23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
Y24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
Y25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
Y26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
Y27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
Y28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
Y29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
Y30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
Y31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
Y32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVI. Propagation delay time (CL to output, V_{DD} = 5 V, t_{PLH} and t_{PHL}) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Z1	A1	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 150 pF, t_{PLH}	35	19	26,27,36,12, 37,25,6	80	800	ns	21,22
Z2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
Z3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
Z4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
Z5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
Z6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
Z7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
Z8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
Z9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
Z10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
Z11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
Z12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
Z13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
Z14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
Z15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
Z16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVI. Propagation delay time (CL to output, V_{DD} = 5 V, t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Z17	A1	Propagation delay time (CL to output), V_{DD} = 5 V, C_L = 150 pF, t_{PHL}	35	19	26,27,36,12, 37,25,6	80	800	ns	21,22
Z18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
Z19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
Z20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
Z21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
Z22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
Z23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
Z24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
Z25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
Z26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
Z27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
Z28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
Z29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
Z30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
Z31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
Z32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVII. Propagation delay time (CL to output, V_{DD} = 10 V, t_{PLH} and t_{PHL}) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
AA1	A1	Propagation delay time (CL to output), V_{DD} = 10 V, C_L = 50 pF, t_{PLH}	35	19	26,27,36,12, 37,25,6	25	250	ns	21,22
AA2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
AA3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
AA4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
AA5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
AA6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
AA7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
AA8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
AA9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
AA10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
AA11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
AA12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
AA13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
AA14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
AA15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
AA16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVII. Propagation delay time (CL to output, V_{DD} = 10 V, t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
AA17	A1	Propagation delay time (CL to output), V_{DD} = 10 V, C_L = 50 pF, t_{PHL}	35	19	26,27,36,12, 37,25,6	25	250	ns	21,22
AA18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
AA19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
AA20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
AA21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
AA22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
AA23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
AA24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
AA25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
AA26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
AA27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
AA28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
AA29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
AA30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
AA31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
AA32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVIII. Propagation delay time (CL to output, V_{DD} = 15 V, t_{PLH} and t_{PHL}) test.

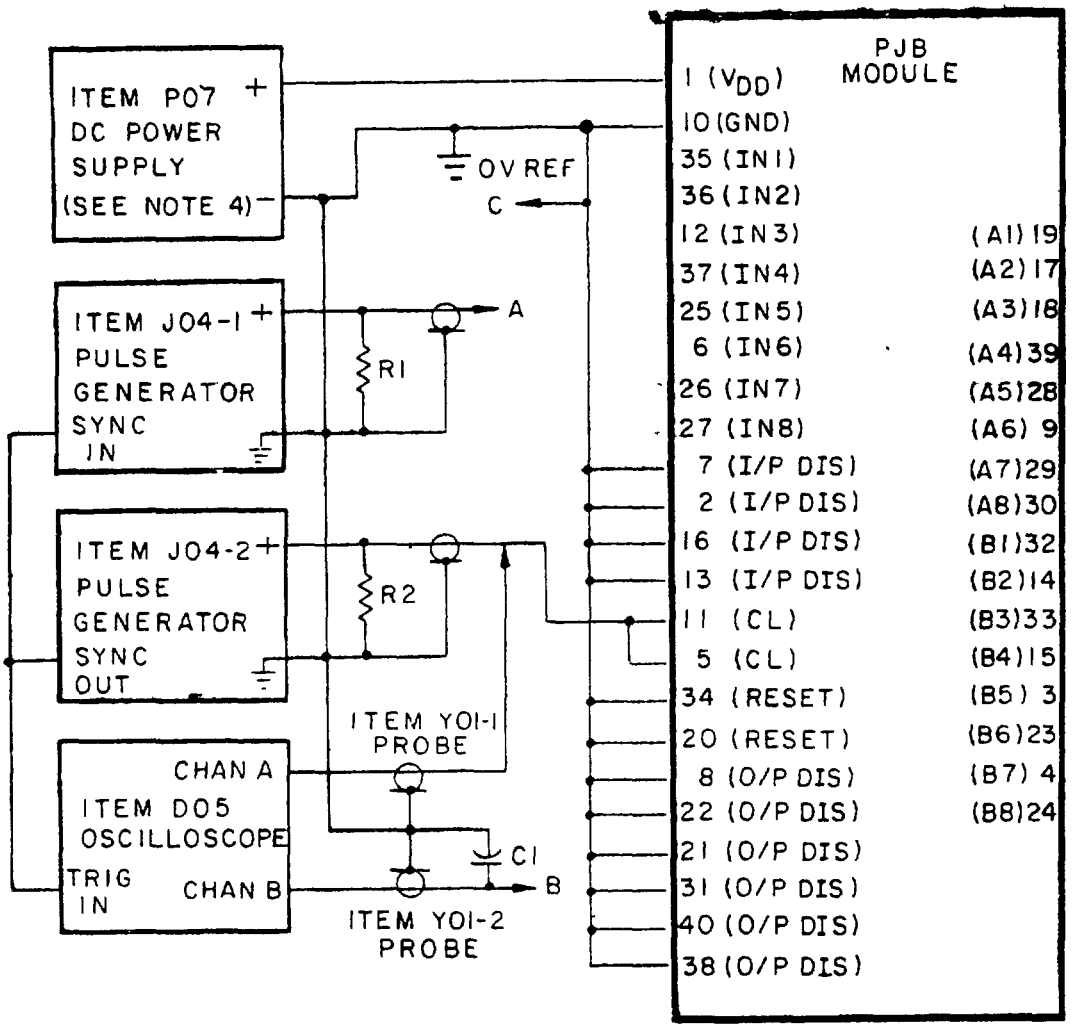
Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
BB1	A1	Propagation delay time (CL to output), V_{DD} = 15 V, C_L = 50 pF, t_{PLH}	35	19	26,27,36,12, 37,25,6	18	180	ns	21,22
BB2	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
BB3	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
BB4	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
BB5	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
BB6	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
BB7	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
BB8	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
BB9	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
BB10	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
BB11	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
BB12	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
BB13	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
BB14	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
BB15	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
BB16	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

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TABLE XXVIII. Propagation delay time (CL to output, V_{DD} = 15 V, t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
BB17	A1	Propagation delay time (CL to output), V_{DD} = 15 V, C_L = 50 pF, t_{PHL}	35	19	26,27,36,12, 37,25,6	18	180	ns	21,22
BB18	B1	"	35	32	26,27,36,12, 37,25,6	"	"	"	"
BB19	A2	"	36	17	26,27,35,12, 37,25,6	"	"	"	"
BB20	B2	"	36	14	26,27,35,12, 37,25,6	"	"	"	"
BB21	A3	"	12	18	26,27,35,36, 37,25,6	"	"	"	"
BB22	B3	"	12	33	26,27,35,36, 37,25,6	"	"	"	"
BB23	A4	"	37	39	26,27,35,36, 12,25,6	"	"	"	"
BB24	B4	"	37	15	26,27,35,36, 12,25,6	"	"	"	"
BB25	A5	"	25	28	26,27,35,36, 12,37,6	"	"	"	"
BB26	B5	"	25	3	26,27,35,36, 12,37,6	"	"	"	"
BB27	A6	"	6	9	26,27,35,36, 12,37,25	"	"	"	"
BB28	B6	"	6	23	26,27,35,36, 12,37,25	"	"	"	"
BB29	A7	"	26	29	6,27,35,36, 12,37,25	"	"	"	"
BB30	B7	"	26	4	6,27,35,36, 12,37,25	"	"	"	"
BB31	A8	"	27	30	6,26,35,36, 12,37,25	"	"	"	"
BB32	B8	"	27	24	6,26,35,36, 12,37,25	"	"	"	"

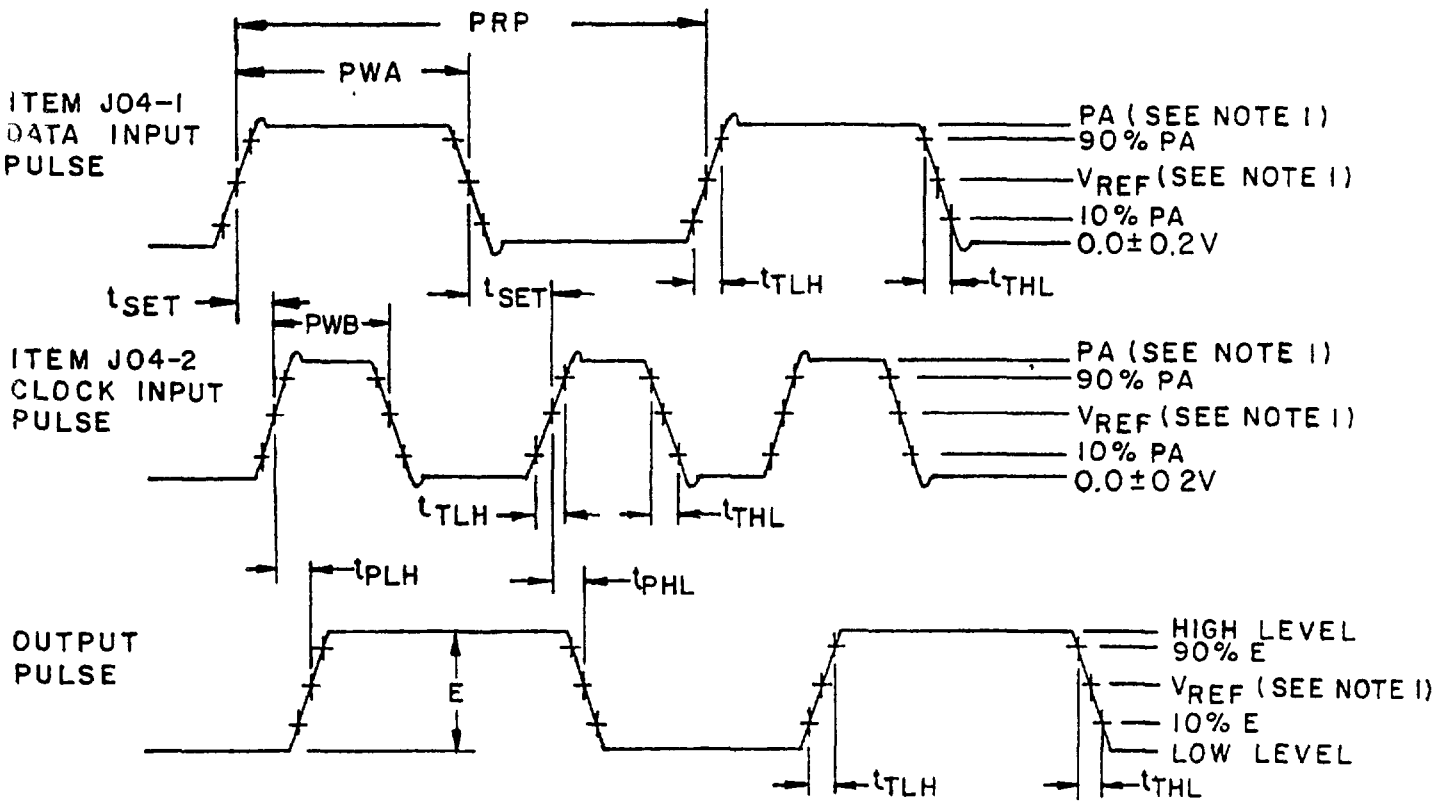
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NOTES:
1. R1 = R2 = 51Ω ±5%, 5 W.
2. C1 = 50 pF ±10% (including probe and parasitic capacitance) for test conditions S1-S16, T1-T16, U1-U16, V1-V16, W1-W16, X1-X16, Y1-Y32, AA1-AA32, and BB1-BB32.
3. C1 = 150 pF ±10% (including probe and parasitic capacitance) for test conditions S17-S32, T17-T32, and Z1-Z32.
   Set item P07 as follows:
   V_{DD} = 5 V ±1% for test conditions S1-S32, T1-T32, Y1-Y32, and Z1-Z32.
   V_{DD} = 10 V ±1% for test conditions U1-U16, V1-V16, and AA1-AA32.
   V_{DD} = 15 V ±1% for test conditions W1-W16, X1-X16, and BB1-BB32.

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (CL to output, t_{PLH} and t_{PHL}) test circuit for tables XIX through XXVIII.

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NOTES:

1. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PRP = 4.0 ±0.4 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

   PWB = 1.0 ±0.1 µs.
   t_{SET} = 250 ±25 ns.

   PA = 5.0 ±0.1 V for test conditions S1-S32, T1-T32, Y1-Y32, and Z1-Z32.
   = 10.0 ±0.1 V for test conditions U1-U16, V1-V16, and AA1-AA32.
   = 15.0 ±0.1 V for test conditions W1-W16, X1-X16, and BB1-BB32.
   V_{REF} = 2.5 ±0.1 V for test conditions S1-S32, T1-T32, Y1-Y32, and Z1-Z32.
   = 5.0 ±0.1 V for test conditions U1-U16, V1-V16, and AA1-AA32.
   = 7.5 ±0.1 V for test conditions W1-W16, X1-X16, and BB1-BB32.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions S1-S32, T1-T32, Y1-Y32, and Z1-Z32.
   ≥ 9.5 V for test conditions U1-U16, V1-V16, and AA1-AA32.
   ≥ 13.5 V for test conditions W1-W16, X1-X16, and BB1-BB32.
   Low level ≤ 0.4 V for test conditions S1-S32, T1-T32, Y1-Y32, and Z1-Z32.
   ≤ 0.5 V for teest conditions U1-U16, V1-V16, and AA1-AA32.
   ≤ 1.5 V for test conditions W1-W16, X1-X16, and BB1-BB32.
3. The lead length of electrial conductors between resistors R1 and R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (CL to output, t_{PLH} and t_{PHL}) test circuit waveforms for tables XIX through XXVIII.

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TABLE XXIX. Propagation delay time (Reset to output, t_{PHL}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
CC1	A1	Propagation delay time (Reset to output), V_{DD} = 5 V, C_L = 50 pF, t_{PHL}	19	60	600	ns	23,24
CC2	A2	"	17	"	"	"	"
CC3	A3	"	18	"	"	"	"
CC4	A4	"	39	"	"	"	"
CC5	A5	"	28	"	"	"	"
CC6	A6	"	9	"	"	"	"
CC7	A7	"	29	"	"	"	"
CC8	A8	"	30	"	"	"	"
CC9	B1	"	32	"	"	"	"
CC10	B2	"	14	"	"	"	"
CC11	B3	"	33	"	"	"	"
CC12	B4	"	15	"	"	"	"
CC13	B5	"	3	"	"	"	"
CC14	B6	"	23	"	"	"	"
CC15	B7	"	4	"	"	"	"
CC16	B8	"	24	"	"	"	"

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TABLE XXIX. Propagation delay time (Reset to output, t_{PHL}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
CC17	A1	Propagation delay time (Reset to output), V_{DD} = 5 V, C_L = 150 pF, t_{PHL}	19	80	800	ns	23,24
CC18	A2	"	17	"	"	"	"
CC19	A3	"	18	"	"	"	"
CC20	A4	"	39	"	"	"	"
CC21	A5	"	28	"	"	"	"
CC22	A6	"	9	"	"	"	"
CC23	A7	"	29	"	"	"	"
CC24	A8	"	30	"	"	"	"
CC25	B1	"	32	"	"	"	"
CC26	B2	"	14	"	"	"	"
CC27	B3	"	33	"	"	"	"
CC28	B4	"	15	"	"	"	"
CC29	B5	"	3	"	"	"	"
CC30	B6	"	23	"	"	"	"
CC31	B7	"	4	"	"	"	"
CC32	B8	"	24	"	"	"	"

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TABLE XXX. Propagation delay time (Reset to output, t_{PHL}) test.

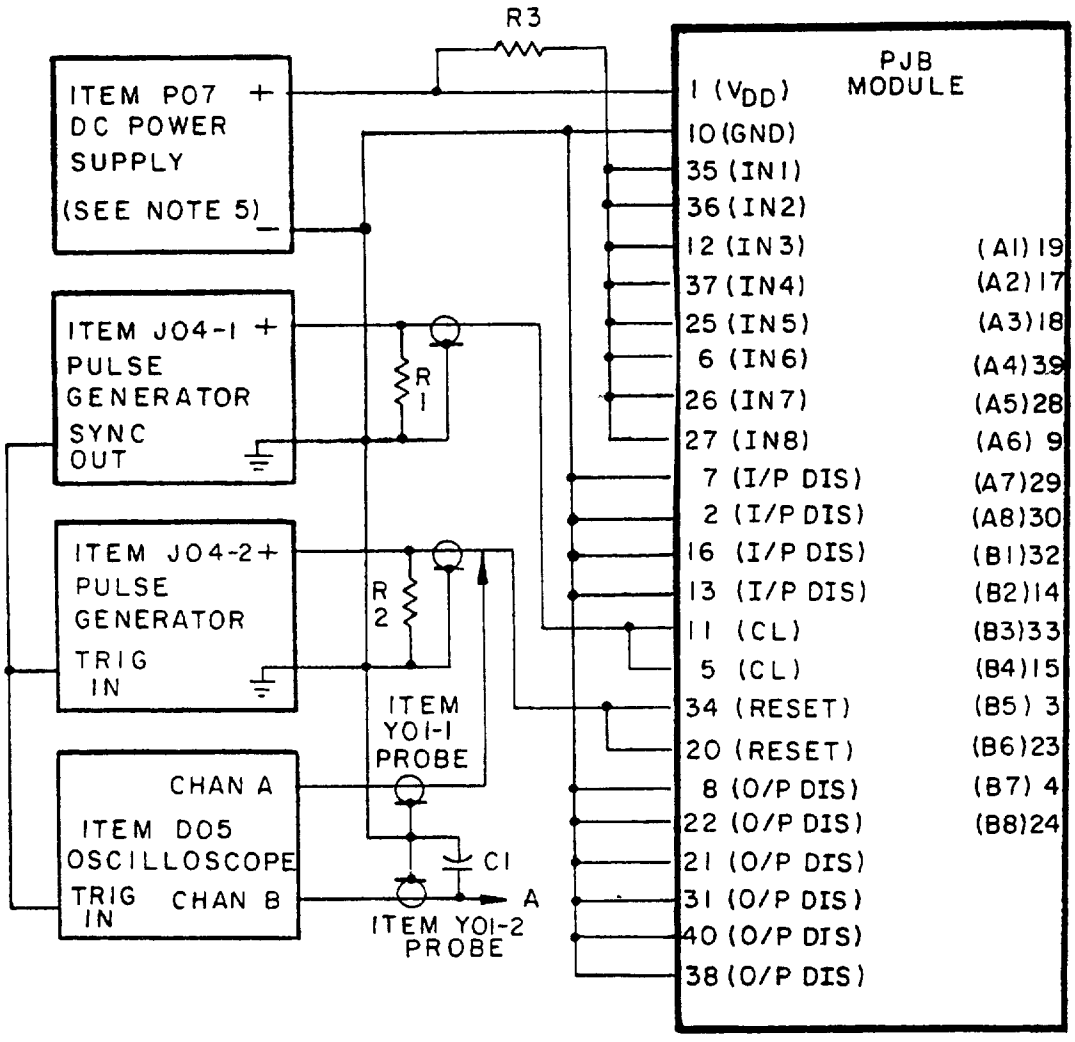
Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
DD1	A1	Propagation delay time (Reset to output), t_{PHL}, V_{DD} = 10 V, C_L = 50 pF	19	25	250	ns	23,24
DD2	A2	"	17	"	"	"	"
DD3	A3	"	18	"	"	"	"
DD4	A4	"	39	"	"	"	"
DD5	A5	"	28	"	"	"	"
DD6	A6	"	9	"	"	"	"
DD7	A7	"	29	"	"	"	"
DD8	A8	"	30	"	"	"	"
DD9	B1	"	32	"	"	"	"
DD10	B2	"	14	"	"	"	"
DD11	B3	"	33	"	"	"	"
DD12	B4	"	15	"	"	"	"
DD13	B5	"	3	"	"	"	"
DD14	B6	"	23	"	"	"	"
DD15	B7	"	4	"	"	"	"
DD16	B8	"	24	"	"	"	"

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TABLE XXXI. Propagation delay time (Reset to output, t_{PHL}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
EE1	A1	Propagation delay time (Reset to output), t_{PHL}, V_{DD} = 15 V, C_L = 50 pF	19	18	180	ns	23,24
EE2	A2	"	17	"	"	"	"
EE3	A3	"	18	"	"	"	"
EE4	A4	"	39	"	"	"	"
EE5	A5	"	28	"	"	"	"
EE6	A6	"	9	"	"	"	"
EE7	A7	"	29	"	"	"	"
EE8	A8	"	30	"	"	"	"
EE9	B1	"	32	"	"	"	"
EE10	B2	"	14	"	"	"	"
EE11	B3	"	33	"	"	"	"
EE12	B4	"	15	"	"	"	"
EE13	B5	"	3	"	"	"	"
EE14	B6	"	23	"	"	"	"
EE15	B7	"	4	"	"	"	"
EE16	B8	"	24	"	"	"	"

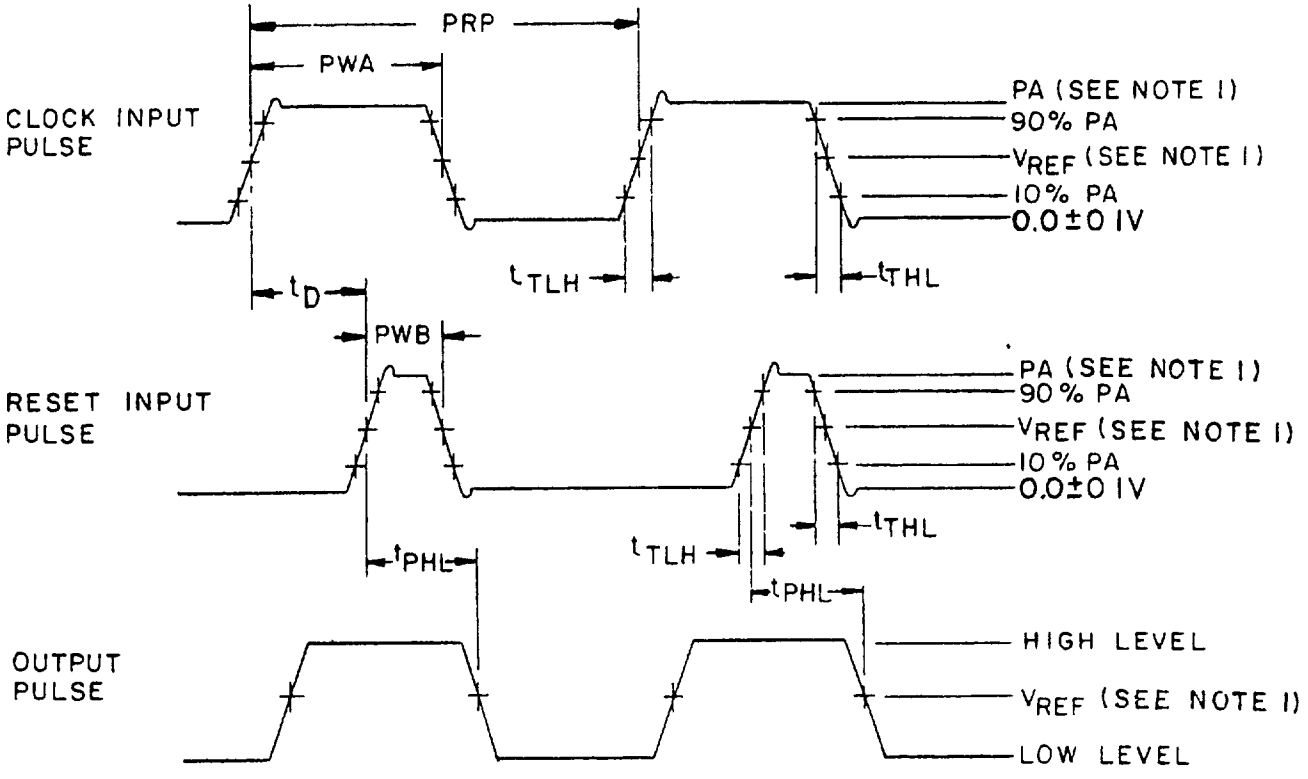
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NOTES:
1. R1 = R2 = 51Ω ±5%, 5 W.
2. R3 = 1.5 kΩ ±1% 1/8 W.
3. C1 = 50 pF ±10% (including probe and parasitic capacitance) for test conditions CC1-CC16, DD1-DD16, and EE1-EE16.
4. C1 = 150 pF ±10% (including probe and parasitic capacitance) for test conditions CC17-CC32.
5. Set item P07 as follows:
   V_{DD} = 5 V ±1% for test conditions CC1-CC32.
   V_{DD} = 10 V ±1% for test conditions DD1-DD16.
   V_{DD} = 15 V ±1% for test conditions EE1-EE16.

FIGURE 23. Propagation delay time (Reset to output, t_{PHL}) test circuit for tables XXIX, XXX, and XXXI.

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NOTES:
1. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PRP = 4.0 ±0.4 µs.
   t_D = 1.5 ±0.1 µs.
   t_{TLH} = t_{THL} = 10 ±2 ns.

   PWB = 370 +30, -0 ns for test conditions CC1-CC32.
   = 150 +15, -0 ns for test conditions DD1-DD16.
   = 110 +11, -0 ns for test conditions EE1-EE16.
   PA = 5.0 ±0.1 V for test conditions CC1-CC32.
   = 10.0 ±0.1 V for test conditions DD1-DD16.
   = 15.0 ±0.1 V for test conditions EE1-EE16.
   V_{REF} = 2.5 ±0.1 V for test conditions CC1-CC32.
   = 5.0 ±0.1 V for test conditions DD1-DD16.
   = 7.5 ±0.1 V for test conditions EE1-EE16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions CC1-CC32.
   ≥ 9.5 V for test conditions DD1-DD16.
   ≥ 13.5 V for test conditions EE1-EE16.
   Low level ≤ 0.4 V for test conditions CC1-CC32.
   ≤ 0.5 V for test conditions DD1-DD16.
   ≤ 1.5 V for test conditions EE1-EE16.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 24. Propagation delay time (Reset to output, t_{PHL}) test circuit waveforms for tables XXIX, XXX, and XXXI.

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TABLE XXXII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
FF1	A1	Disable delay time (t_{PLZ}), V_{DD} = 5 V, C_L = 50 pF	19	70	700	ns	25,26
FF2	A2	"	17	"	"	"	"
FF3	A3	"	18	"	"	"	"
FF4	A4	"	39	"	"	"	"
FF5	A5	"	28	"	"	"	"
FF6	A6	"	9	"	"	"	"
FF7	A7	"	29	"	"	"	"
FF8	A8	"	30	"	"	"	"
FF9	B1	"	32	"	"	"	"
FF10	B2	"	14	"	"	"	"
FF11	B3	"	33	"	"	"	"
FF12	B4	"	15	"	"	"	"
FF13	B5	"	3	"	"	"	"
FF14	B6	"	23	"	"	"	"
FF15	B7	"	4	"	"	"	"
FF16	B8	"	24	"	"	"	"

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TABLE XXXII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
FF17	A1 Enable delay time (t_{PZL}), V_{DD} = 5 V, C_L = 50 pF	19	17	170	ns	25,26
FF18	A2 "	17	"	"	"	"
FF19	A3 "	18	"	"	"	"
FF20	A4 "	39	"	"	"	"
FF21	A5 "	28	"	"	"	"
FF22	A6 "	9	"	"	"	"
FF23	A7 "	29	"	"	"	"
FF24	A8 "	30	"	"	"	"
FF25	B1 "	32	"	"	"	"
FF26	B2 "	14	"	"	"	"
FF27	B3 "	33	"	"	"	"
FF28	B4 "	15	"	"	"	"
FF29	B5 "	3	"	"	"	"
FF30	B6 "	23	"	"	"	"
FF31	B7 "	4	"	"	"	"
FF32	B8 "	24	"	"	"	"

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TABLE XXXII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
FF33	A1	Enable delay time (t_{PHZ}), V_{DD} = 5 V, C_L = 50 pF	19	0.15	1.50	µs	25,26
FF34	A2	"	17	"	"	"	"
FF35	A3	"	18	"	"	"	"
FF36	A4	"	39	"	"	"	"
FF37	A5	"	28	"	"	"	"
FF38	A6	"	9	"	"	"	"
FF39	A7	"	29	"	"	"	"
FF40	A8	"	30	"	"	"	"
FF41	B1	"	32	"	"	"	"
FF42	B2	"	14	"	"	"	"
FF43	B3	"	33	"	"	"	"
FF44	B4	"	15	"	"	"	"
FF45	B5	"	3	"	"	"	"
FF46	B6	"	23	"	"	"	"
FF47	B7	"	4	"	"	"	"
FF48	B8	"	24	"	"	"	"

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TABLE XXXII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
FF49	A1	Fnable delay time (t_{PZH}), V_{DD} = 5 V, C_L = 50 pF	19	14	140	ns	25,26
FF50	A2	"	17	"	"	"	"
FF51	A3	"	18	"	"	"	"
FF52	A4	"	39	"	"	"	"
FF53	A5	"	28	"	"	"	"
FF54	A6	"	9	"	"	"	"
FF55	A7	"	29	"	"	"	"
FF56	A8	"	30	"	"	"	"
FF57	B1	"	32	"	"	"	"
FF58	B2	"	14	"	"	"	"
FF59	B3	"	33	"	"	"	"
FF60	B4	"	15	"	"	"	"
FF61	B5	"	3	"	"	"	"
FF62	B6	"	23	"	"	"	"
FF63	B7	"	4	"	"	"	"
FF64	B8	"	24	"	"	"	"

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TABLE XXXIII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
GG1	A1	Disable delay time (t_{PLZ}), V_{DD} = 10 V, C_L = 50 pF	19	66	660	ns	25,26
GG2	A2	"	17	"	"	"	"
GG3	A3	"	18	"	"	"	"
GG4	A4	"	39	"	"	"	"
GG5	A5	"	28	"	"	"	"
GG6	A6	"	9	"	"	"	"
GG7	A7	"	29	"	"	"	"
GG8	A8	"	30	"	"	"	"
GG9	B1	"	32	"	"	"	"
GG10	B2	"	14	"	"	"	"
GG11	B3	"	33	"	"	"	"
GG12	B4	"	15	"	"	"	"
GG13	B5	"	3	"	"	"	"
GG14	B6	"	23	"	"	"	"
GG15	B7	"	4	"	"	"	"
GG16	B8	"	24	"	"	"	"

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TABLE XXXIII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
GG17	A1	Enable delay time (t_{PZL}), V_{DD} = 10 V, C_L = 50 pF	19	6	60	ns	25,26
GG18	A2	"	17	"	"	"	"
GG19	A3	"	18	"	"	"	"
GG20	A4	"	39	"	"	"	"
GG21	A5	"	28	"	"	"	"
GG22	A6	"	9	"	"	"	"
GG23	A7	"	29	"	"	"	"
GG24	A8	"	30	"	"	"	"
GG25	B1	"	32	"	"	"	"
GG26	B2	"	14	"	"	"	"
GG27	B3	"	33	"	"	"	"
GG28	B4	"	15	"	"	"	"
GG29	B5	"	3	"	"	"	"
GG30	B6	"	23	"	"	"	"
GG31	B7	"	4	"	"	"	"
GG32	B8	"	24	"	"	"	"

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TABLE XXXIII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
GG33	A1	Enable delay time (t_{PHZ}), V_{DD} = 10 V, C_L = 50 pF	19	0.15	1.50	µs	25,26
GG34	A2	"	17	"	"	"	"
GG35	A3	"	18	"	"	"	"
GG36	A4	"	39	"	"	"	"
GG37	A5	"	28	"	"	"	"
GG38	A6	"	9	"	"	"	"
GG39	A7	"	29	"	"	"	"
GG40	A8	"	30	"	"	"	"
GG41	B1	"	32	"	"	"	"
GG42	B2	"	14	"	"	"	"
GG43	B3	"	33	"	"	"	"
GG44	B4	"	15	"	"	"	"
GG45	B5	"	3	"	"	"	"
GG46	B6	"	23	"	"	"	"
GG47	B7	"	4	"	"	"	"
GG48	B8	"	24	"	"	"	"

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TABLE XXXIII. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
GG49	A1	Enable delay time (t_{PZH}), V_{DD} = 10 V, C_L = 50 pF	19	6	60	ns	25,26
GG50	A2	"	17	"	"	"	"
GG51	A3	"	18	"	"	"	"
GG52	A4	"	39	"	"	"	"
GG53	A5	"	28	"	"	"	"
GG54	A6	"	9	"	"	"	"
GG55	A7	"	29	"	"	"	"
GG56	A8	"	30	"	"	"	"
GG57	B1	"	32	"	"	"	"
GG58	B2	"	14	"	"	"	"
GG59	B3	"	33	"	"	"	"
GG60	B4	"	15	"	"	"	"
GG61	B5	"	3	"	"	"	"
GG62	B6	"	23	"	"	"	"
GG63	B7	"	4	"	"	"	"
GG64	B8	"	24	"	"	"	"

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TABLE XXXIV. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
HH1	A1	Disable delay time (t_{PLZ}), V_{DD} = 15 V, C_L = 50 pF	19	62	620	ns	25,26
HH2	A2	"	17	"	"	"	"
HH3	A3	"	18	"	"	"	"
HH4	A4	"	39	"	"	"	"
HH5	A5	"	28	"	"	"	"
HH6	A6	"	9	"	"	"	"
HH7	A7	"	29	"	"	"	"
HH8	A8	"	30	"	"	"	"
HH9	B1	"	32	"	"	"	"
HH10	B2	"	14	"	"	"	"
HH11	B3	"	33	"	"	"	"
HH12	B4	"	15	"	"	"	"
HH13	B5	"	3	"	"	"	"
HH14	B6	"	23	"	"	"	"
HH15	B7	"	4	"	"	"	"
HH16	B8	"	24	"	"	"	"

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TABLE XXXIV. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
HH17	A1	Enable delay time (t_{PZL}), V_{DD} = 15 V, C_L = 50 pF	19	5	50	ns	25,26
HH18	A2	"	17	"	"	"	"
HH19	A3	"	18	"	"	"	"
HH20	A4	"	39	"	"	"	"
HH21	A5	"	28	"	"	"	"
HH22	A6	"	9	"	"	"	"
HH23	A7	"	29	"	"	"	"
HH24	A8	"	30	"	"	"	"
HH25	B1	"	32	"	"	"	"
HH26	B2	"	14	"	"	"	"
HH27	B3	"	33	"	"	"	"
HH28	B4	"	15	"	"	"	"
HH29	B5	"	3	"	"	"	"
HH30	B6	"	23	"	"	"	"
HH31	B7	"	4	"	"	"	"
HH32	B8	"	24	"	"	"	"

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TABLE XXXIV. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ} and t_{PZH}) test - Continued.

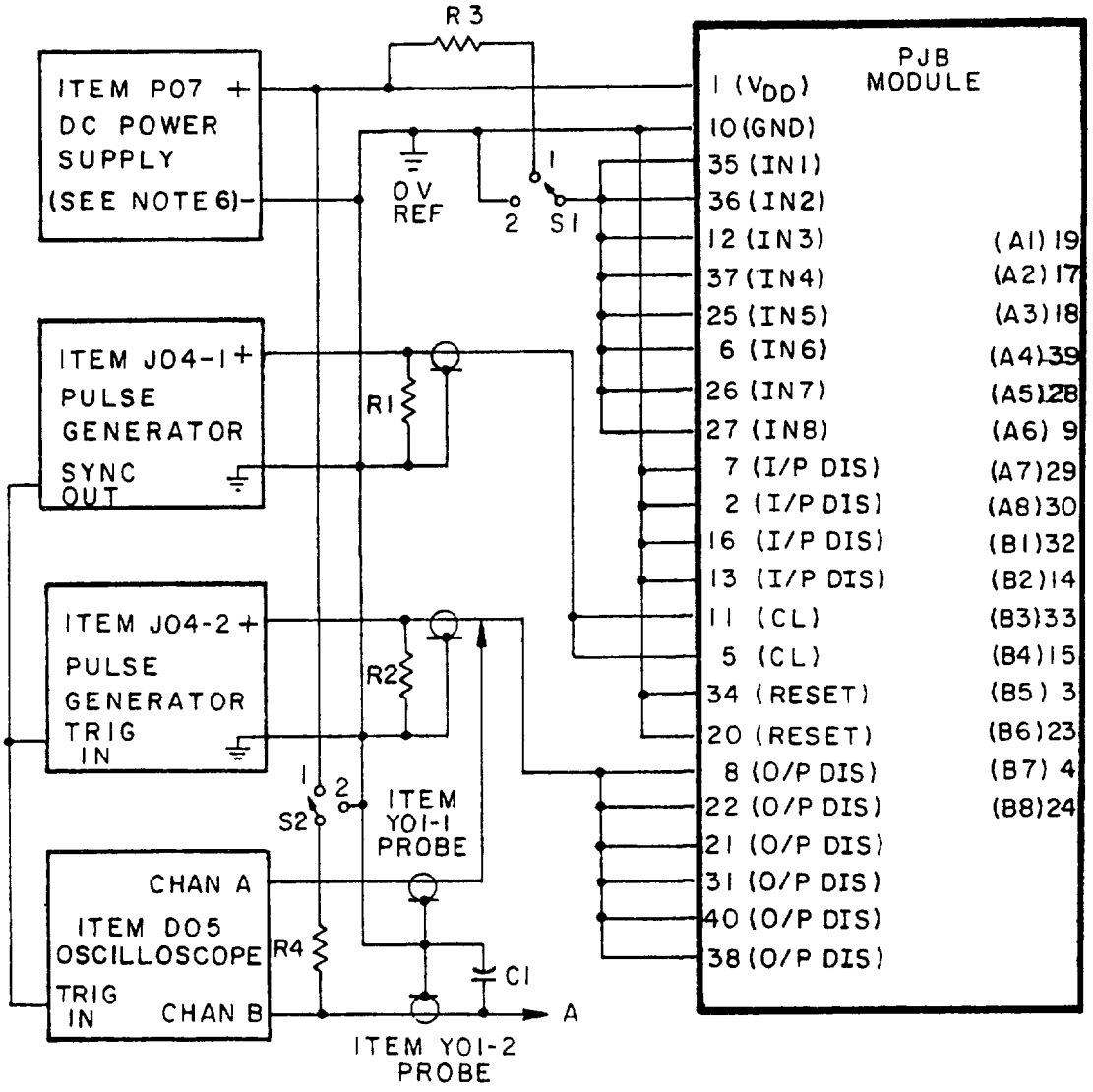
Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
HH33	A1	Disable delay time (t_{PHZ}), V_{DD} = 15 V, C_L = 50 pF	19	0.14	1.40	µs	25,26
HH34	A2	"	17	"	"	"	"
HH35	A3	"	18	"	"	"	"
HH36	A4	"	39	"	"	"	"
HH37	A5	"	28	"	"	"	"
HH38	A6	"	9	"	"	"	"
HH39	A7	"	29	"	"	"	"
HH40	A8	"	30	"	"	"	"
HH41	B1	"	32	"	"	"	"
HH42	B2	"	14	"	"	"	"
HH43	B3	"	33	"	"	"	"
HH44	B4	"	15	"	"	"	"
HH45	B5	"	3	"	"	"	"
HH46	B6	"	23	"	"	"	"
HH47	B7	"	4	"	"	"	"
HH48	B8	"	24	"	"	"	"

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TABLE XXXIV. Disable/Enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, and t_{PZH}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
HH49	A1	Enable delay time (t_{PZH}), V_{DD} = 15 V, C_L = 50 pF	19	5	45	ns	25,26
HH50	A2	"	17	"	"	"	"
HH51	A3	"	18	"	"	"	"
HH52	A4	"	39	"	"	"	"
HH53	A5	"	28	"	"	"	"
HH54	A6	"	9	"	"	"	"
HH55	A7	"	29	"	"	"	"
HH56	A8	"	30	"	"	"	"
HH57	B1	"	32	"	"	"	"
HH58	B2	"	14	"	"	"	"
HH59	B3	"	33	"	"	"	"
HH60	B4	"	15	"	"	"	"
HH61	B5	"	3	"	"	"	"
HH62	B6	"	23	"	"	"	"
HH63	B7	"	4	"	"	"	"
HH64	B8	"	24	"	"	"	"

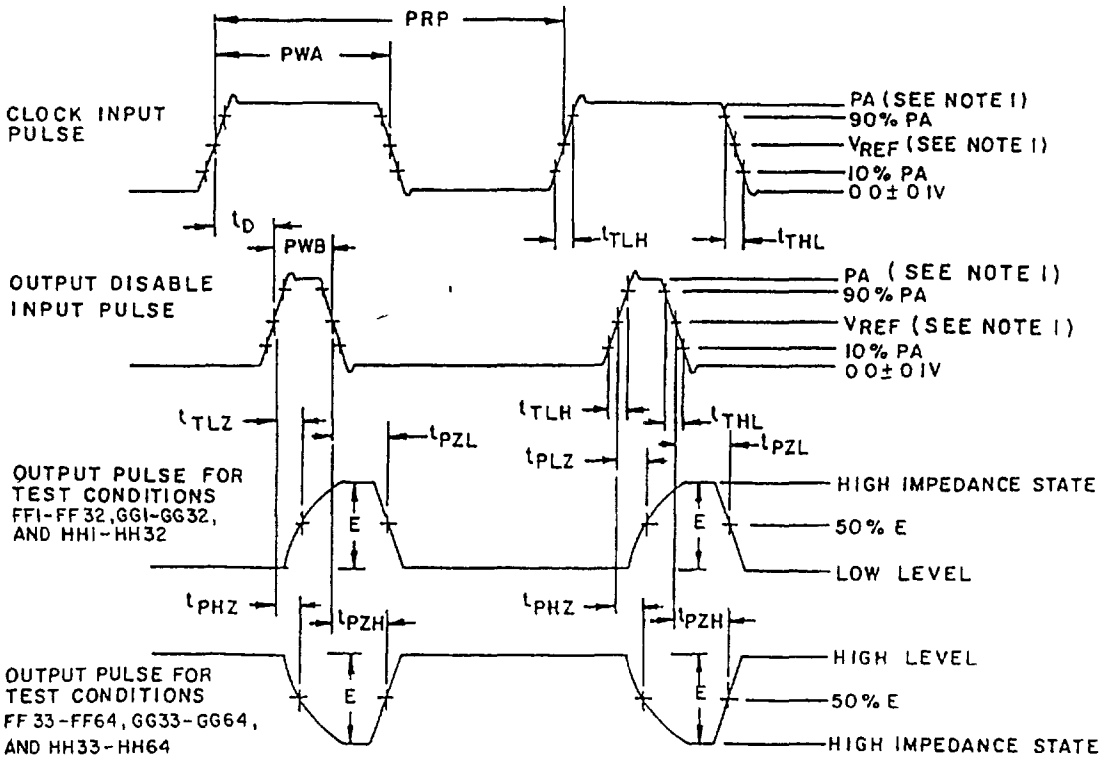
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NOTES:
1. R1 = R2 = 51Ω ±5%, 5 W.
2. R3 = 1.5 kΩ ±5%, 1/8 W.
3. R4 = 12.78 kΩ ±1%, 1/8 W for t_{PZL} and t_{PLZ}.
   R4 = 32.86 kΩ ±1%, 1/8 W for t_{PZH} and t_{PHZ}.
4. C1 = 50 pF ±10% (including probe and parasitic capacitance).
5. S1 = S2 = SPDT switch.
6. Set item P07 as follows:
   V_{DD} = 5 V ±1% for test conditions FF1-FF64.
   V_{DD} = 10 V ±1% for test conditions GG1-GG64.
   V_{DD} = 15 V ±1% for test conditions HH1-HH64.
7. For test conditions FF1-FF32, GG1-GG32, and HH1-HH32 place switch S1 in position 2 and switch S2 in position 1.
8. For test conditions FF33-FF64, GG33-GG64, and HH33-HH64 place switch S1 in position 1 and S2 in position 2.

* FIGURE 25. Disable/enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, t_{PZH}) test circuit for tables XXXII, XXXIII, and XXXIV.

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NOTES:
1. Input pulse parameters:
   PWA = 7.50 ±0.75 µs.
   PRP = 15.0 ±1.5 µs.
   PWB = 5.0 ±0.5 µs.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_D = 1.50 \pm 0.15$ µs.

   PA = 5.0 ±0.1 V for test conditions FF1-FF64.
      = 10.0 ±0.1 V for test conditions GG1-GG64.
      = 15.0 ±0.1 V for test conditions HH1-HH64.
   V_{REF} = 2.5 ±0.1 V for test conditions FF1-FF64.
      = 5.0 ±0.1 V for test conditions GG1-GG64.
      = 7.5 ±0.1 V for test conditions HH1-HH64.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level $\geq$ 4.6 V for test conditions FF1-FF32.
   $\geq$ 9.5 V for test conditions GG1-GG32.
   $\geq$ 13.5 V for test conditions HH1-HH32.
   Low level $\leq$ 0.4 V for test conditions FF33-FF64.
   $\leq$ 0.5 V for test conditions GG33-GG64.
   $\leq$ 1.5 V for test conditions HH33-HH64.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes YO1-1 and YO1-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 26. Disable/enable delay time (t_{PLZ}, t_{PZL}, t_{PHZ}, t_{PZH}) test circuit waveforms for tables XXXII, XXXIII, and XXXIV.

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TABLE XXXV. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
JJ1	A1	Hold time $t_{H(H)}$ (V_{DD} = 5 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
JJ2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
JJ3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
JJ4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
JJ5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
JJ6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
JJ7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
JJ8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
JJ9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
JJ10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
JJ11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
JJ12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
JJ13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
JJ14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
JJ15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
JJ16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXV. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
JJ17	A1	Hold time $t_{H(L)}$ (V_{DD} = 5 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
JJ18	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
JJ19	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
JJ20	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
JJ21	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
JJ22	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
JJ23	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
JJ24	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
JJ25	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
JJ26	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
JJ27	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
JJ28	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
JJ29	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
JJ30	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
JJ31	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
JJ32	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVI. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
KK1	A1	Setup time $t_{S(H)}$ (V_{DD} = 5 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
KK2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
KK3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
KK4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
KK5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
KK6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
KK7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
KK8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
KK9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
KK10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
KK11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
KK12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
KK13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
KK14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
KK15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
KK16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVI. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
KK17	A1	Setup time $t_{S(L)}$ (V_{DD} = 5 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
KK18	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
KK19	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
KK20	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
KK21	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
KK22	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
KK23	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
KK24	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
KK25	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
KK26	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
KK27	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
KK28	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
KK29	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
KK30	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
KK31	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
KK32	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVII. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
LL1	A1	Hold time $t_{H(H)}$ (V_{DD} = 10 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
LL2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
LL3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
LL4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
LL5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
LL6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
LL7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
LL8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
LL9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
LL10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
LL11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
LL12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
LL13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
LL14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
LL15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
LL16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVII. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
LL17	A1	Hold time $t_{H(L)}$ (V_{DD} = 10 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
LL18	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
LL19	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
LL20	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
LL21	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
LL22	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
LL23	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
LL24	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
LL25	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
LL26	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
LL27	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
LL28	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
LL29	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
LL30	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
LL31	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
LL32	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVIII. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
MM1	A1	Setup time $t_{S(H)}$ (V_{DD} = 10 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
MM2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
MM3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
MM4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
MM5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
MM6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
MM7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
MM8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
MM9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
MM10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
MM11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
MM12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
MM13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
MM14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
MM15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
MM16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXVIII. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
MM17	A1	Setup time $t_{S(L)}$ (V_{DD} = 10 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
MM18	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
MM19	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
MM20	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
MM21	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
MM22	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
MM23	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
MM24	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
MM25	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
MM26	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
MM27	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
MM28	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
MM29	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
MM30	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
MM31	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
MM32	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXIX. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
NN1	A1	Hold time $t_{H(H)}$ (V_{DD} = 15 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
NN2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
NN3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
NN4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
NN5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
NN6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
NN7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
NN8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
NN9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
NN10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
NN11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
NN12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
NN13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
NN14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
NN15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
NN16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XXXIX. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
NN17	A1 Hold time $t_{H(L)}$ (V_{DD} = 15 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,28
NN18	B1 "	35	36,12,37, 25,6,26,27	32	"	"	"	"
NN19	A2 "	36	35,12,37, 25,6,26,27	17	"	"	"	"
NN20	B2 "	36	35,12,37, 25,6,26,27	14	"	"	"	"
NN21	A3 "	12	35,36,37, 25,6,26,27	18	"	"	"	"
NN22	B3 "	12	35,36,37, 25,6,26,27	33	"	"	"	"
NN23	A4 "	37	35,36,12, 25,6,26,27	39	"	"	"	"
NN24	B4 "	37	35,36,12, 25,6,26,27	15	"	"	"	"
NN25	A5 "	25	35,36,12, 37,6,26,27	28	"	"	"	"
NN26	B5 "	25	35,36,12, 37,6,26,27	3	"	"	"	"
NN27	A6 "	6	35,36,12, 37,25,26,27	9	"	"	"	"
NN28	B6 "	6	35,36,12, 37,25,26,27	23	"	"	"	"
NN29	A7 "	26	35,36,12, 37,25,6,27	29	"	"	"	"
NN30	B7 "	26	35,36,12, 37,25,6,27	4	"	"	"	"
NN31	A8 "	27	35,36,12, 37,25,6,26	30	"	"	"	"
NN32	B8 "	27	35,36,12, 37,25,6,26	24	"	"	"	"

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TABLE XL. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test.

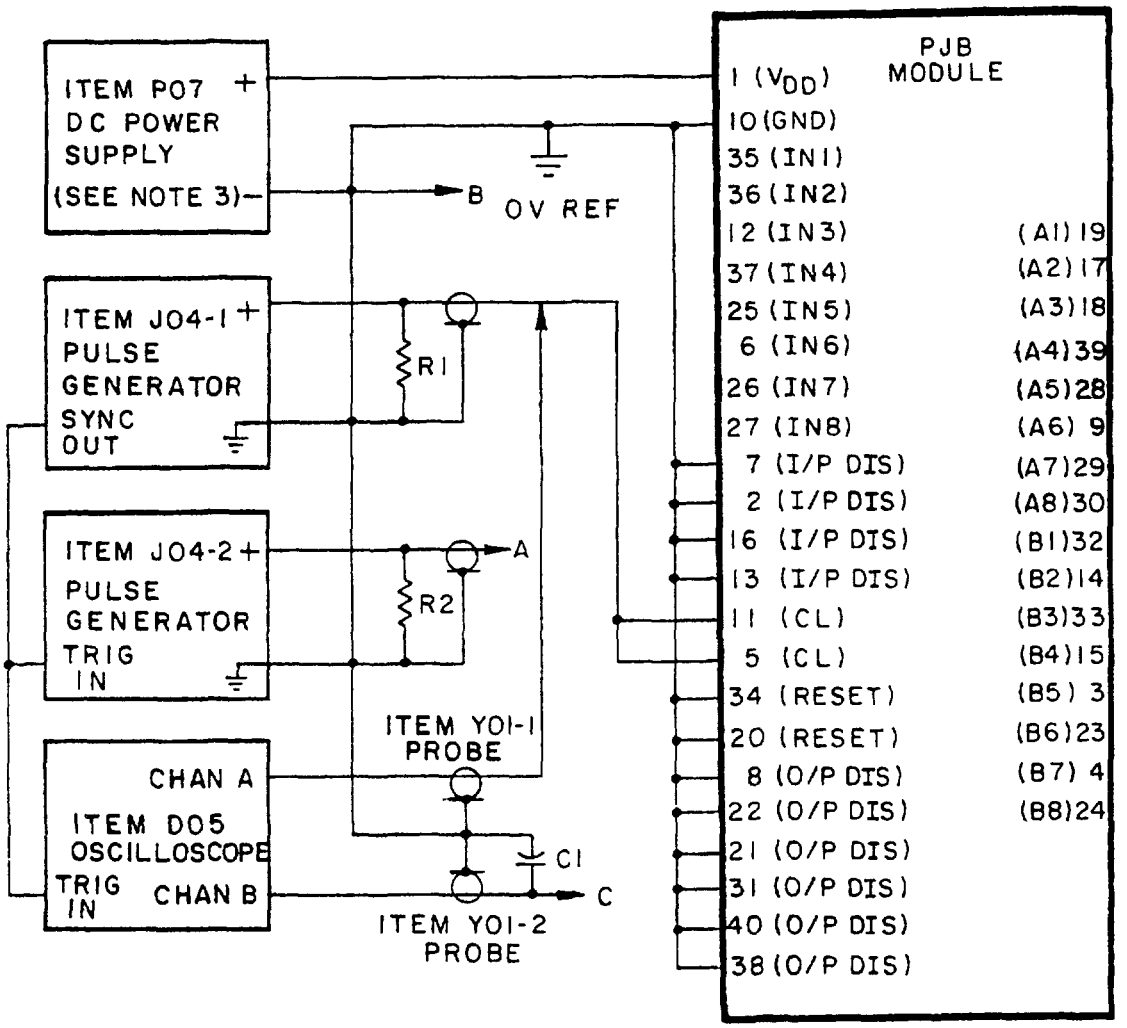
Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
PP1	A1	Setup time $t_{S(H)}$ (V_{DD} = 15 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
PP2	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
PP3	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
PP4	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
PP5	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
PP6	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
PP7	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
PP8	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
PP9	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
PP10	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
PP11	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
PP12	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
PP13	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
PP14	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
PP15	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
PP16	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

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Check the source to verify that this is the current version before use.

TABLE XL. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
PP17	A1	Setup time $t_{S(L)}$ (V_{DD} = 15 V), C_L = 50 pF	35	36,12,37, 25,6,26,27	19	3	5	µs	27,29
PP18	B1	"	35	36,12,37, 25,6,26,27	32	"	"	"	"
PP19	A2	"	36	35,12,37, 25,6,26,27	17	"	"	"	"
PP20	B2	"	36	35,12,37, 25,6,26,27	14	"	"	"	"
PP21	A3	"	12	35,36,37, 25,6,26,27	18	"	"	"	"
PP22	B3	"	12	35,36,37, 25,6,26,27	33	"	"	"	"
PP23	A4	"	37	35,36,12, 25,6,26,27	39	"	"	"	"
PP24	B4	"	37	35,36,12, 25,6,26,27	15	"	"	"	"
PP25	A5	"	25	35,36,12, 37,6,26,27	28	"	"	"	"
PP26	B5	"	25	35,36,12, 37,6,26,27	3	"	"	"	"
PP27	A6	"	6	35,36,12, 37,25,26,27	9	"	"	"	"
PP28	B6	"	6	35,36,12, 37,25,26,27	23	"	"	"	"
PP29	A7	"	26	35,36,12, 37,25,6,27	29	"	"	"	"
PP30	B7	"	26	35,36,12, 37,25,6,27	4	"	"	"	"
PP31	A8	"	27	35,36,12, 37,25,6,26	30	"	"	"	"
PP32	B8	"	27	35,36,12, 37,25,6,26	24	"	"	"	"

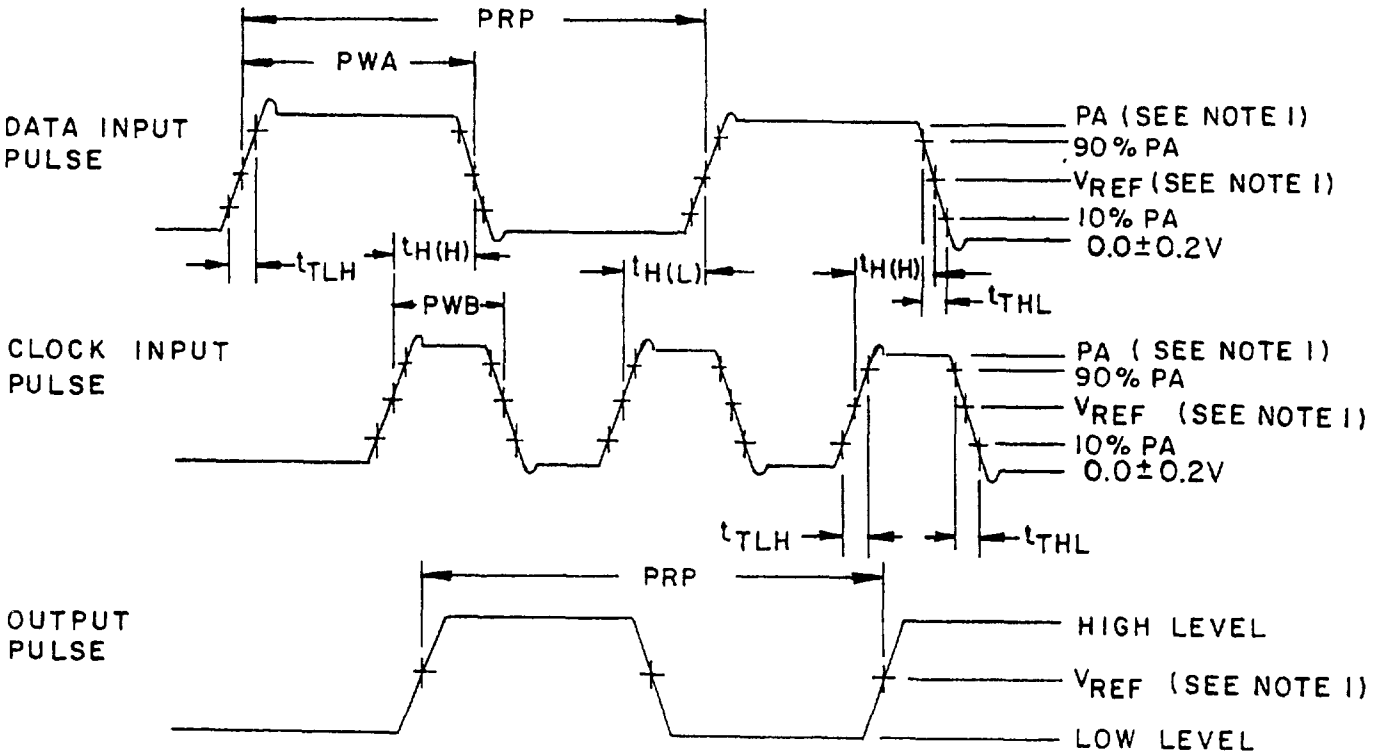
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NOTES:
1. R1 = R2 = 51Ω ±5 W.
2. C1 = 50 pF ±10% (including probe and parasitic capacitance).
3. Set item P07 as follows:
   V_{DD} = 5 V ±1% for test conditions JJ1-JJ32 and KK1-KK32.
   V_{DD} = 10 V ±1% for test conditions LL1-LL32 and MM1-MM32.
   V_{DD} = 15 V ±1% for test conditions NN1-NN32 and PP1-PP32.

FIGURE 27. Hold time ($t_{H(H)}$ and $t_{H(L)}$) and setup time ($t_{S(H)}$ and $t_{S(L)}$) test circuit for tables XXXV through XL.

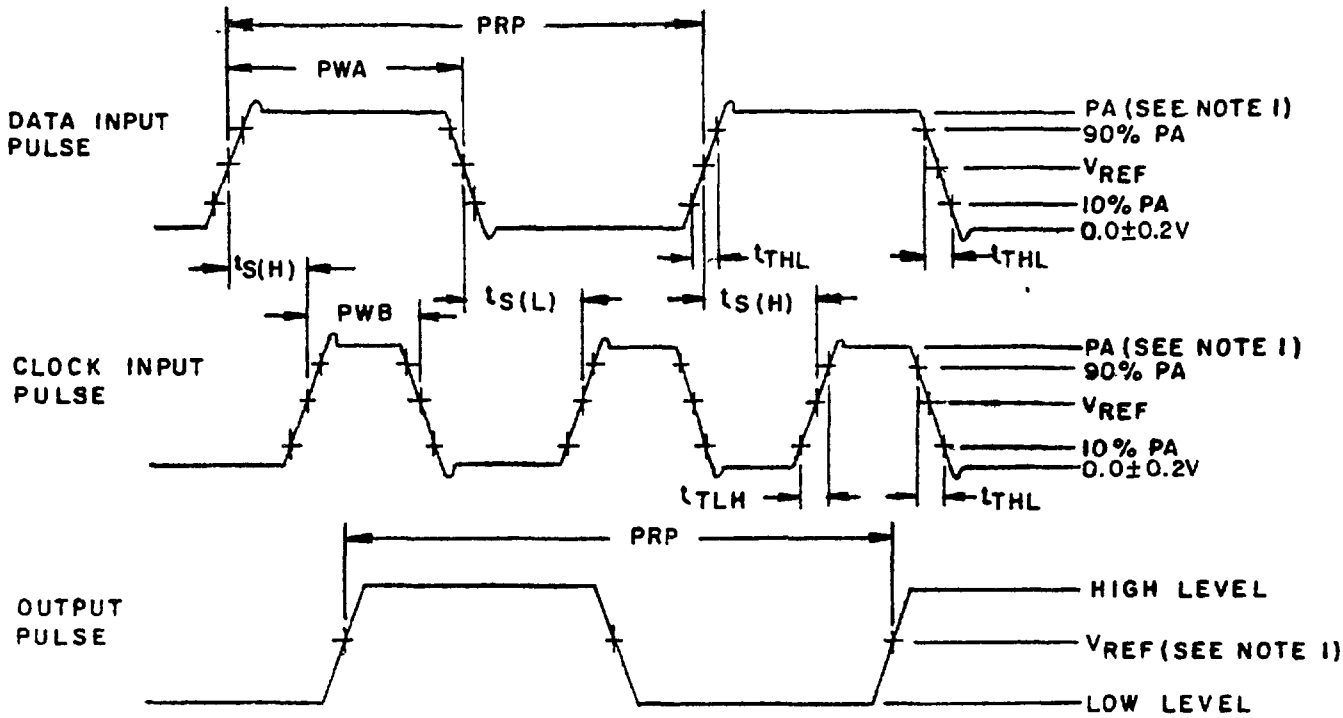
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NOTES:
1. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PWB = 1.0 ±0.1 µs.
   PRP = 4.0 ±0.4 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   $t_{H(H)}$ = 130 +13, -0 ns for test conditions JJ1-JJ16.
   = 200 ±20 ns for test conditions JJ17-JJ32, LL17-LL32, and NN17-NN32.
   = 60 +6, -0 ns for test conditions LL1-LL16.
   = 50 +5, -0 ns for test conditions NN1-NN16.
   $t_{H(L)}$ = 200 ±20 ns for test conditions JJ1-JJ16, LL1-LL16, and NN1-NN16.
   = 130 +13, -0 ns for test conditions JJ17-JJ32.
   = 60 +6, -0 ns for test conditions LL17-LL32.
   = 50 +5, -0 ns for test conditions NN17-NN32.
   PA = 5 ±0.1 V for test conditions JJ1-JJ32.
   = 10.0 ±0.1 V for test conditions LL1-LL32.
   = 15.0 ±0.1 V for test conditions NN1-NN32.
   V_{REF} = 2.5 ±0.1 V for test conditions JJ1-JJ32.
   = 5.0 ±0.1 V for test conditions LL1-LL32.
   = 7.5 ±0.1 V for test conditions NN1-NN32.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions JJ1-JJ32.
   ≥ 9.5 V for test conditions LL1-LL32.
   ≥ 13.5 V for test conditions NN1-NN32.
   Low level ≤ 0.4 V for test conditions JJ1-JJ32.
   ≤ 0.5 V for test conditions LL1-LL32.
   ≤ 1.5 V for test conditions NN1-NN32.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 28. Hold time ($t_{H(H)}$ and $t_{H(L)}$) test circuit waveforms for tables XXXV, XXXVII, and XXXIX.

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NOTES:
1. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PWB = 1.0 ±0.1 µs.
   PRP = 4.0 ±0.4 µs.
   t_{TLH} = t_{THL} = 10 ±2 ns.
   $t_{S(H)}$ = 200 +20, -0 ns for test conditions KK1-KK16.
   = 250 ±25 ns for test conditions KK17-KK32.
   = 80 +8, -0 ns for test conditions MM1-MM16.
   = 60 +6, -0 ns for test conditions PP1-PP16.
   = 200 +20 ns for test conditions MM17-MM32 and PP17-PP32.
   $t_{S(L)}$ = 250 ±25 ns for test conditions KK1-KK16.
   = 200 +20, -0 ns for test conditions KK17-KK32.
   = 200 ±20 ns for test conditions MM1-MM16 and PP1-PP16.
   = 80 +8, -0 ns for test conditions MM17-MM32.
   = 60 +6, -0 ns for test conditions PP17-PP32.
   PA = 5.0 ±0.1 V for test conditions KK1-KK32.
   = 10.0 ±0.1 V for test conditions MM1-MM32.
   = 15.0 ±0.1 V for test conditions PP1-PP32.
   V_{REF} = 2.5 ±0.1 V for test conditions KK1-KK32.
   = 5.0 ±0.1 V for test conditions MM1-MM32.
   = 7.5 ±0.1 V for test conditions PP1-PP32.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level $\geq$ 4.6 V for test conditions KK1-KK32.
   $\geq$ 9.5 V for test conditions MM1-MM32.
   $\geq$ 13.5 V for test conditions PP1-PP32.
   Low level $\leq$ 0.4 V for test conditions KK1-KK32.
   $\leq$ 0.5 V for test conditions MM1-MM32.
   $\leq$ 1.5 V for test conditions PP1-PP32.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

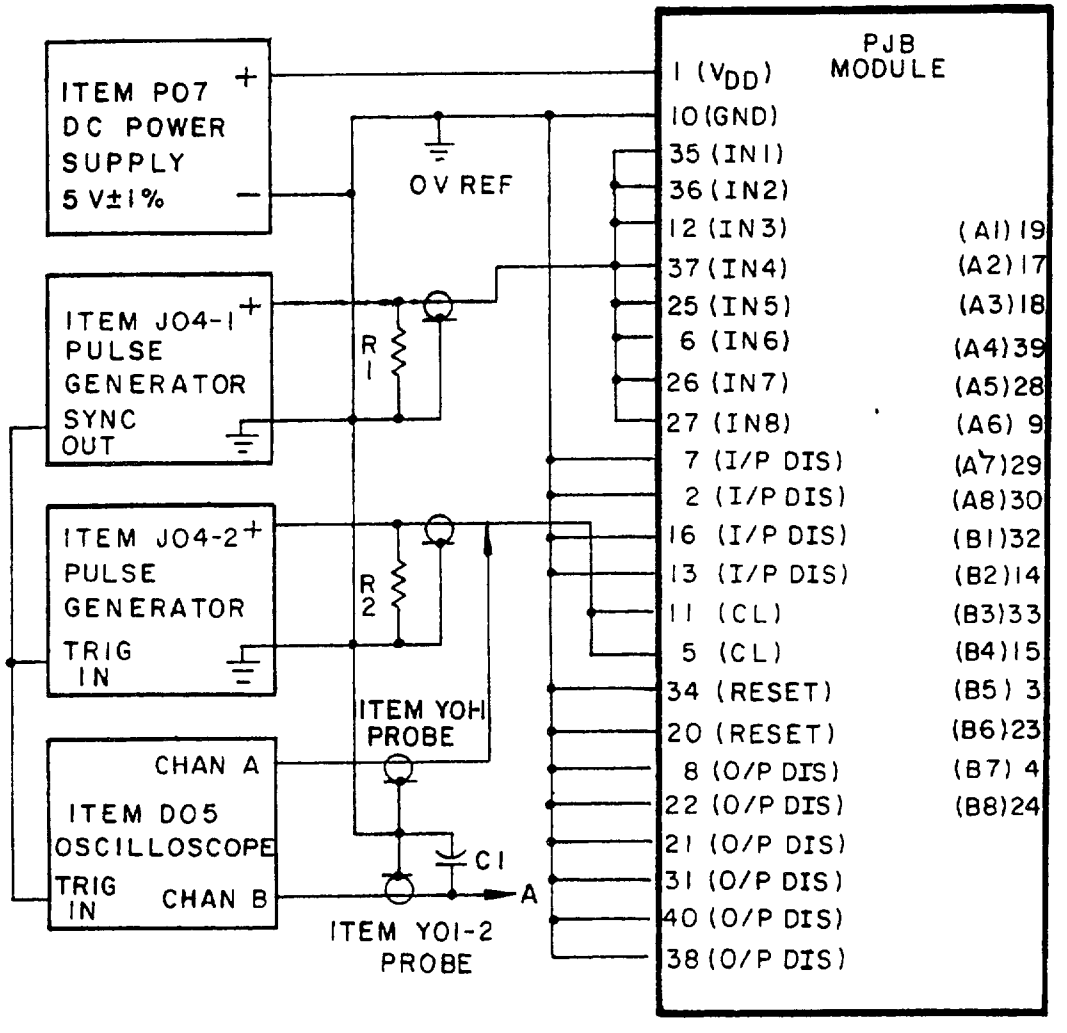
FIGURE 29. Setup time ($t_{S(H)}$ and $t_{S(L)}$) test circuit waveforms for tables XXXVI, XXXVIII, and XL.

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TABLE XLI. Maximum shift frequency check test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
QQ1	A1	Maximum shift frequency check	19	950	1150	ns	30,31
QQ2	A2	"	17	"	"	"	"
QQ3	A3	"	18	"	"	"	"
QQ4	A4	"	39	"	"	"	"
QQ5	A5	"	28	"	"	"	"
QQ6	A6	"	9	"	"	"	"
QQ7	A7	"	29	"	"	"	"
QQ8	A8	"	30	"	"	"	"
QQ9	A5	"	32	"	"	"	"
QQ10	B5	"	14	"	"	"	"
QQ11	A6	"	33	"	"	"	"
QQ12	B6	"	15	"	"	"	"
QQ13	A7	"	3	"	"	"	"
QQ14	B7	"	23	"	"	"	"
QQ15	A8	"	4	"	"	"	"
QQ16	B8	"	24	"	"	"	"

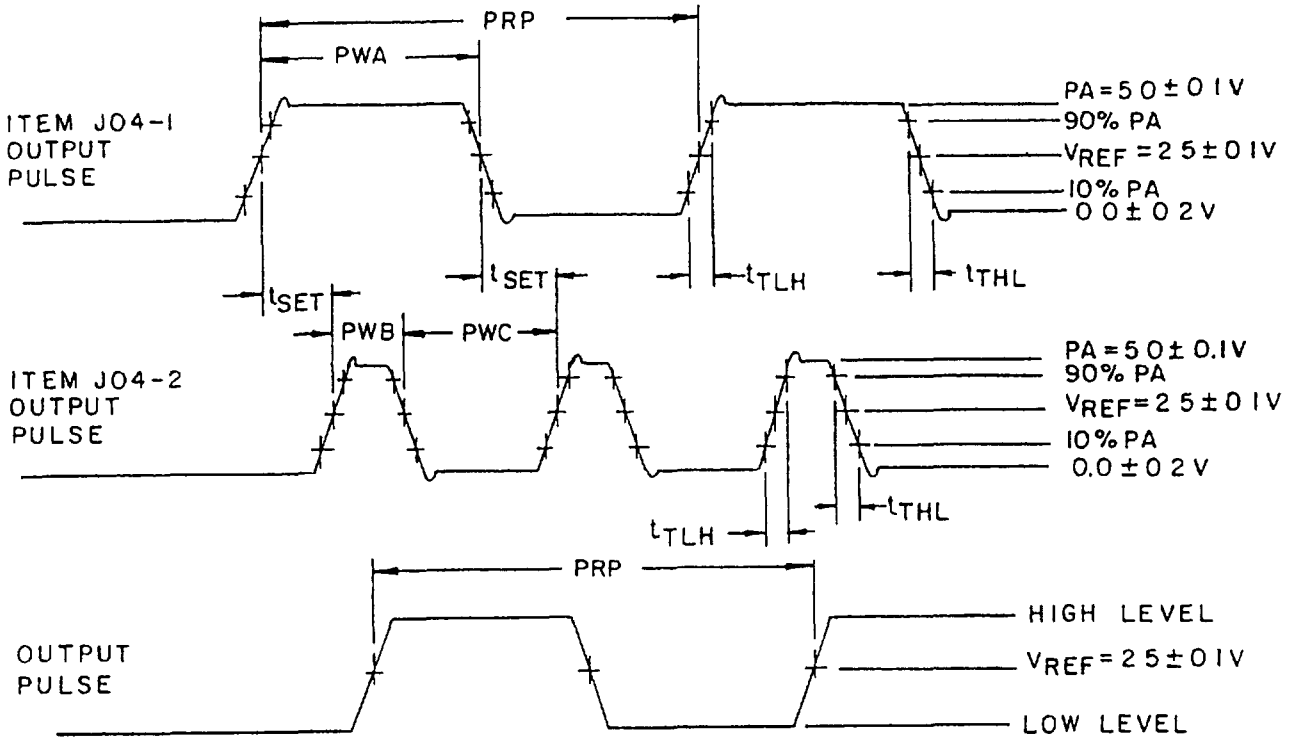
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NOTES:
1. R1 = R2 = 51Ω ±5%, 5 W.
2. C1 = 50 pF ±10% (including probe and parasitic capacitance).

FIGURE 30. Maximum shift frequency check test cicuit for table XLI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
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NOTES:
1. Input pulse parameters:
   PWA = 555 ±20 ns.
   PRP = 1,110 ±40 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{SET} = 250 \pm 25$ ns.

   PWB = PWC = 277 +20, -0 ns.
2. Before making the measurement specified, verify that the output high level is > 4.6 V and the output low level is < 0.4 V.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 31. Maximum shift frequency check test circuit waveforms for table XLI.

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TABLE XLII. Power supply filter capacitance test.

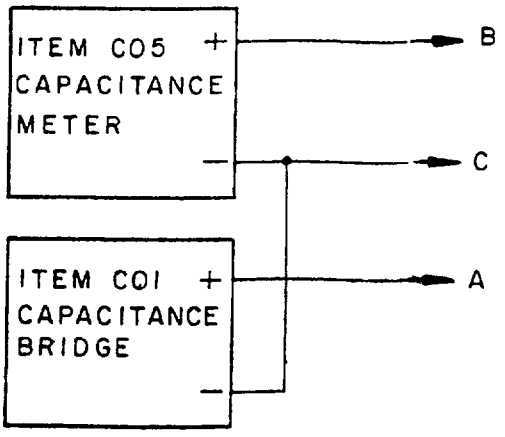
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
RR	Power supply filter capacitance	1	Open	10	70	130	µF	32

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Check the source to verify that this is the current version before use.

TABLE XLIII. Input capacitance test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
SS1	IN1	Input capacitance	Open	35	10	2	15	pF	32
SS2	IN2	"	"	36	"	"	"	"	"
SS3	IN3	"	"	12	"	"	"	"	"
SS4	IN4	"	"	37	"	"	"	"	"
SS5	IN5	"	"	25	"	"	"	"	"
SS6	IN6	"	"	6	"	"	"	"	"
SS7	IN7	"	"	26	"	"	"	"	"
SS8	IN8	"	"	27	"	"	"	"	"
SS9	I/P DIS	"	"	7	"	"	25	"	"
SS10	I/P DIS	"	"	2	"	"	"	"	"
SS11	I/P DIS	"	"	16	"	"	"	"	"
SS12	I/P DIS	"	"	13	"	"	"	"	"
SS13	O/P DIS	"	"	22	"	"	"	"	"
SS14	O/P DIS	"	"	31	"	"	"	"	"
SS15	O/P DIS	"	"	38	"	"	"	"	"
SS16	O/P DIS	"	"	8	"	"	"	"	"
SS17	O/P DIS	"	"	40	"	"	50	"	"
SS18	O/P DIS	"	"	21	"	"	"	"	"
SS19	CL	"	"	11	"	"	"	"	"
SS20	CL	"	"	5	"	"	"	"	"
SS21	RESET	"	"	20	"	"	"	"	"
SS22	RESET	"	"	34	"	"	"	"	"

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PJB MODULE	
1 (V_{DD})	
10 (GND)	
35 (IN1)	
36 (IN2)	
12 (IN3)	(A1) 19
37 (IN4)	(A2) 17
25 (IN5)	(A3) 18
6 (IN6)	(A4) 39
26 (IN7)	(A5) 28
27 (IN8)	(A6) 9
7 (I/P DIS)	(A7) 29
2 (I/P DIS)	(A8) 30
16 (I/P DIS)	(B1) 32
13 (I/P DIS)	(B2) 14
11 (CL)	(B3) 33
5 (CL)	(B4) 15
34 (RESET)	(B5) 3
20 (RESET)	(B6) 23
8 (O/P DIS)	(B7) 4
22 (O/P DIS)	(B8) 24
21 (O/P DIS)	
31 (O/P DIS)	
40 (O/P DIS)	
38 (O/P DIS)	

NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than the pins under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup: Frequency = 1.0 kHz. V_{pk} = 0.3 V maximum

FIGURE 32. Power supply filter capacitance and input capacitance test circuit for tables XLII and XLIII.

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